Python II:

Data Analysis in Python

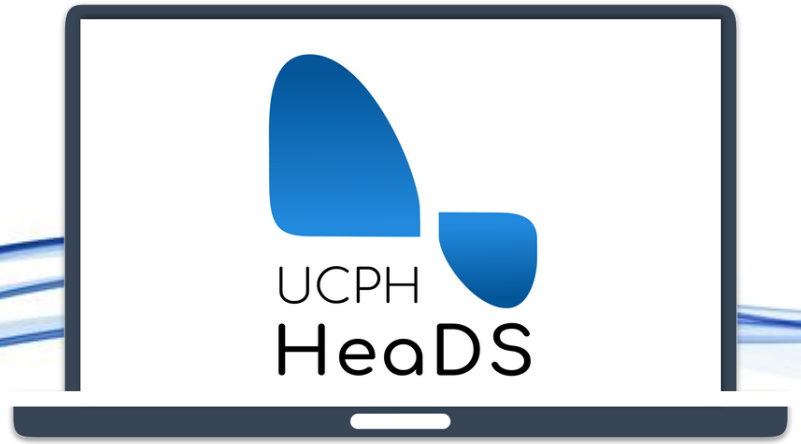


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Center for Health Data Science (HeaDS)

The mission of the Center is to strengthen **health data science & AI at SUND**:

- Active and visible hub for Health Data Science & AI
- Providing research support SUND employees
- Workshops and training environments to improve data science and AI-literacy skills
- Support a network of researchers and educators



TEACHERS



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Special Consultant,
Research IT



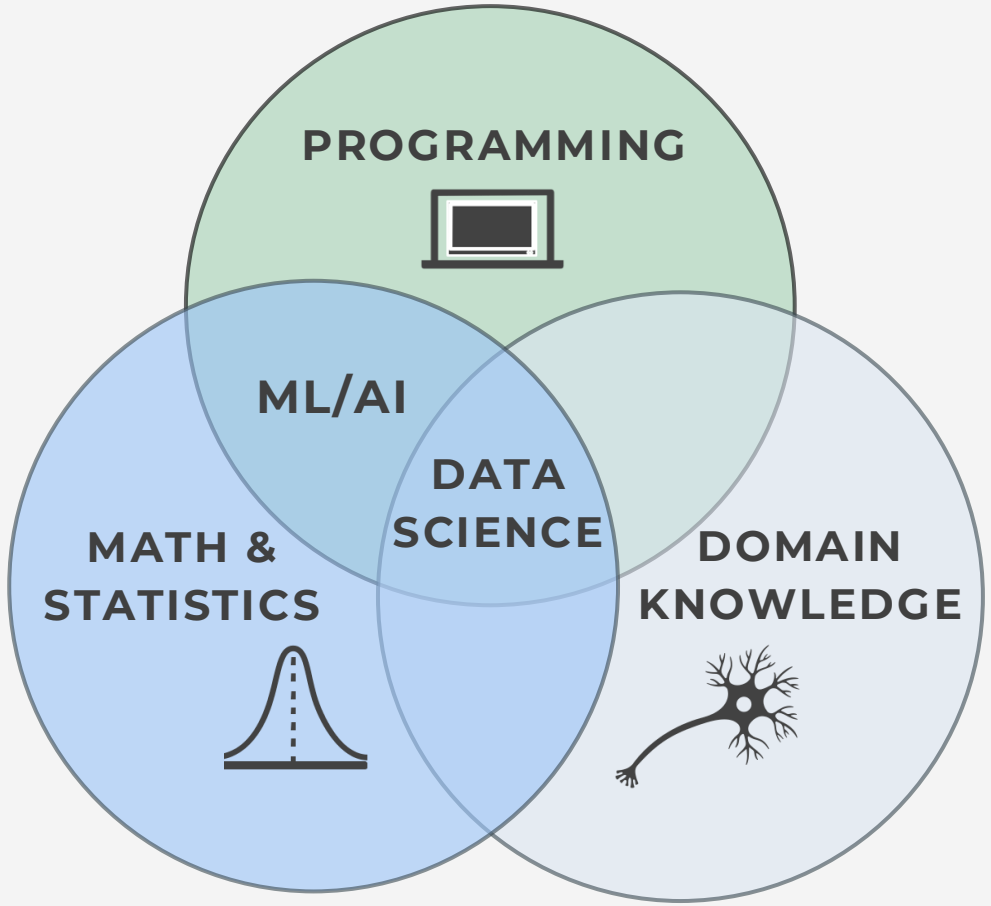
Suze Roostee
Data Scientist,
HeaDS, HDSAI

COURSE PROGRAM

Day 1		Day 2		Day 3	
Coffee					
09:00	Intro	09:00	Functions	09:00	Documentation and Testing
09:15	Numpy and Pandas	10:30	Intro to Modelling: Linear Regression	09:50	K-means Clustering
11:00	Exploratory Data Analysis			11:10	Classes
Lunch					
13:00	PCA	13:00	Linear Regression	13:00	Classes continued
15:15	Classes, attributes & methods	15:30	Classification with Decision Trees	14:00	Project work

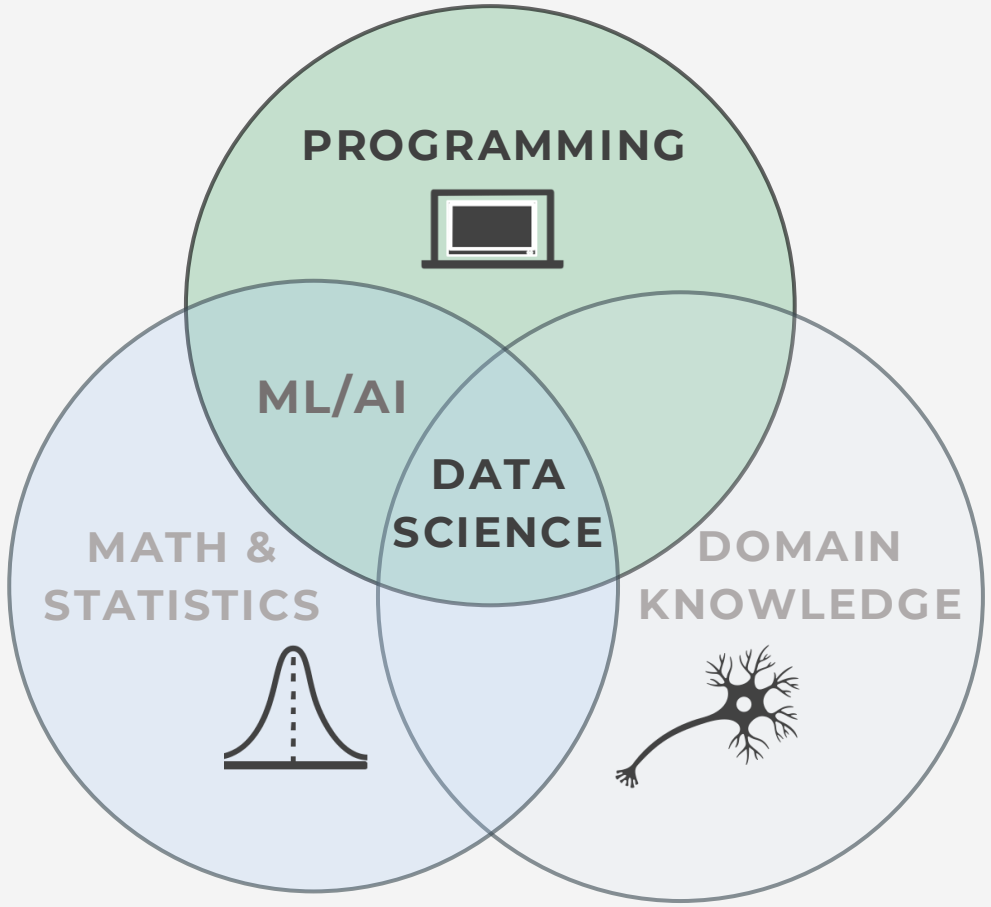
WHAT IS DATA SCIENCE?

- A **cross-disciplinary** undertaking that draws on many disciplines and is in turn becoming part of many disciplines.
- Data matters, one size analysis does not fit all.



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WHAT IS DATA SCIENCE?



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THE DATA'S JOURNEY



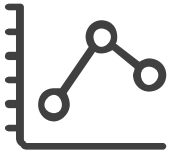
DATA COLLECTION



EXPLORATORY DATA ANALYSIS



MODELLING



MODEL EVALUATION





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BUT FIRST ...



We need some foundations in python to be able to start looking at these things.











DATA WRANGLING PACKAGES

These packages help us to:

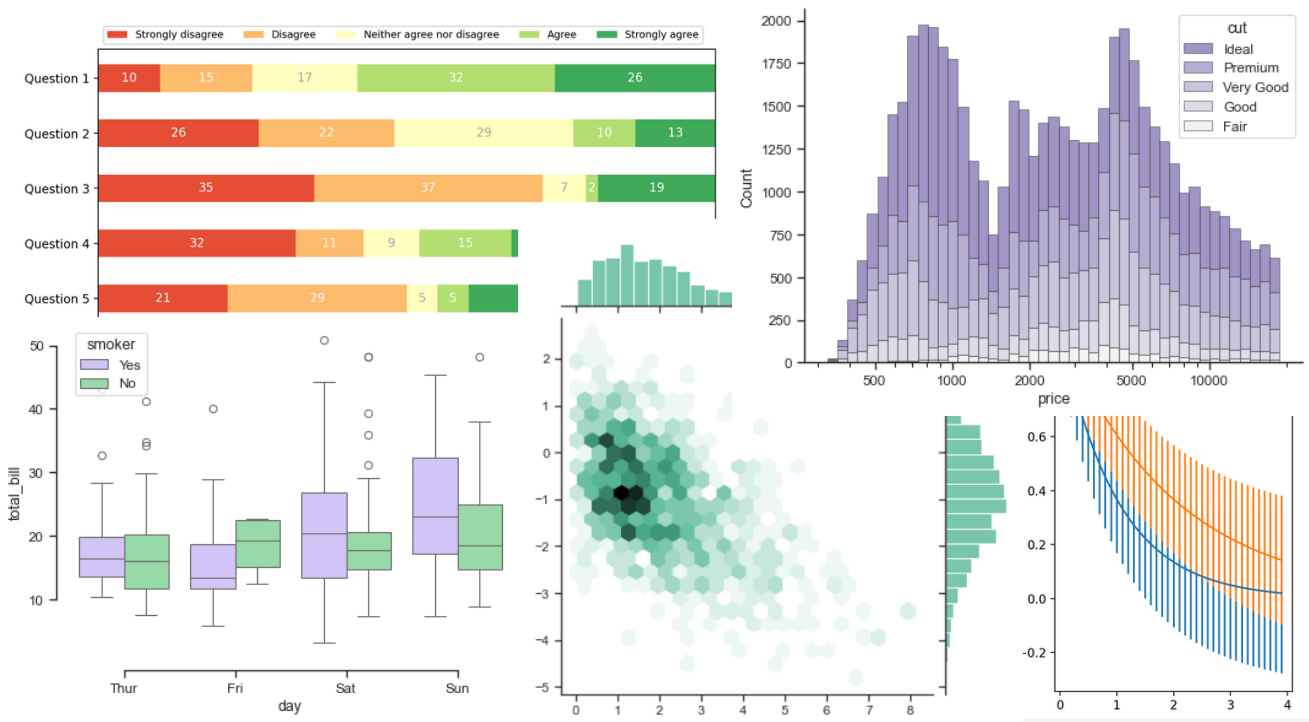
- Represent data in python objects (array and dataframe)
- Load and save data
- Manipulate data
- Investigate data
- Make summaries of data





PLOTTING PACKAGES

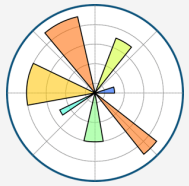
These packages help us to visualize data:



seaborn

matplotlib

MATPLOTLIB



The **seaborn** package is built on top of matplotlib, which is the original plotting library of python.

Advantages

- you can control every aspect of the plot

Drawbacks

- Granular syntax
- Need to define everything yourself
- Inconsistencies in functions and options between plot types

Seaborn is an interface to matplotlib which circumvents many of these problems but at the cost of being less customizable.

DATA SCIENCE PACKAGE

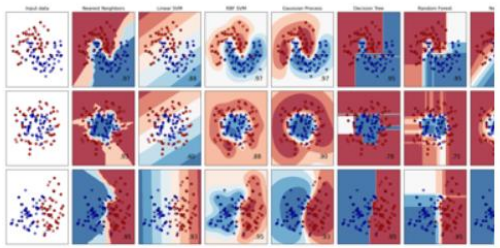
This package helps us to run data science algorithms.



Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.
Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...

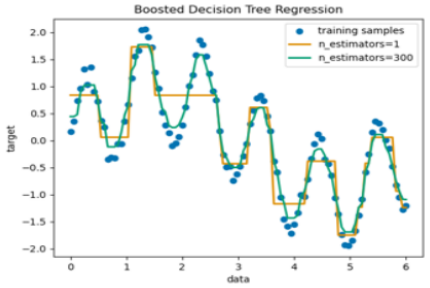


Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.
Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...

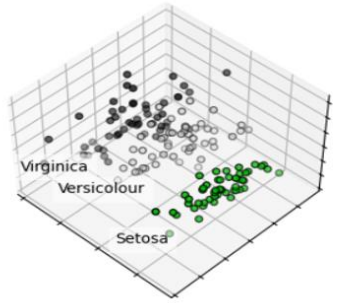


Examples

Dimensionality reduction

Reducing the number of random variables to consider.

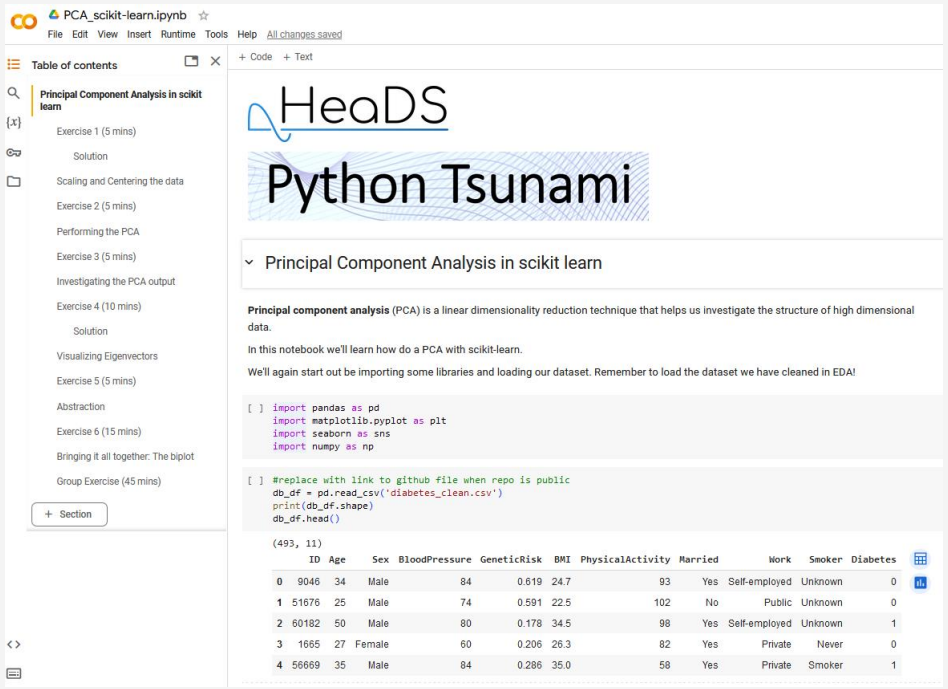
Applications: Visualization, Increased efficiency
Algorithms: PCA, feature selection, non-negative matrix factorization, and more...



Examples

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RUNNING PYTHON



Google Colab

- Jupyter Notebooks executed on Google's servers, not your own machine
- Still browser-based
- Requires no setup and provides 'free' access to computing resources on Google's servers including GPUs
- Connected to a Google account and data and notebooks can be accessed through Google Drive

ACCESSING COURSE MATERIALS

Access the notebooks for this course via GitHub.

You should **copy the notebooks to your google drive** IF you want to save the code you write in this course!

We will do this together for the first notebook.

Python part 2

The links in the schedule will bring you directly to colab. Use Ctrl+click/Cmd+click to open in a new tab.

DAY 1	DAY 2	DAY 3
Morning coffee (optional)	Morning coffee (optional)	Morning coffee (optional)
Intro	Functions	Documentation
Numpy and Pandas		Testing
Exploratory Data Analysis	Linear Regression	Clustering
		Classes
Lunch	Lunch	Lunch
PCA	Categorical Vars	Classes continued
Objects, classes, methods	Classification	Heart failure project (classification) SF birthweight project (regression)
Q & A	Q & A	Q & A

Numpy and Pandas Notebooks

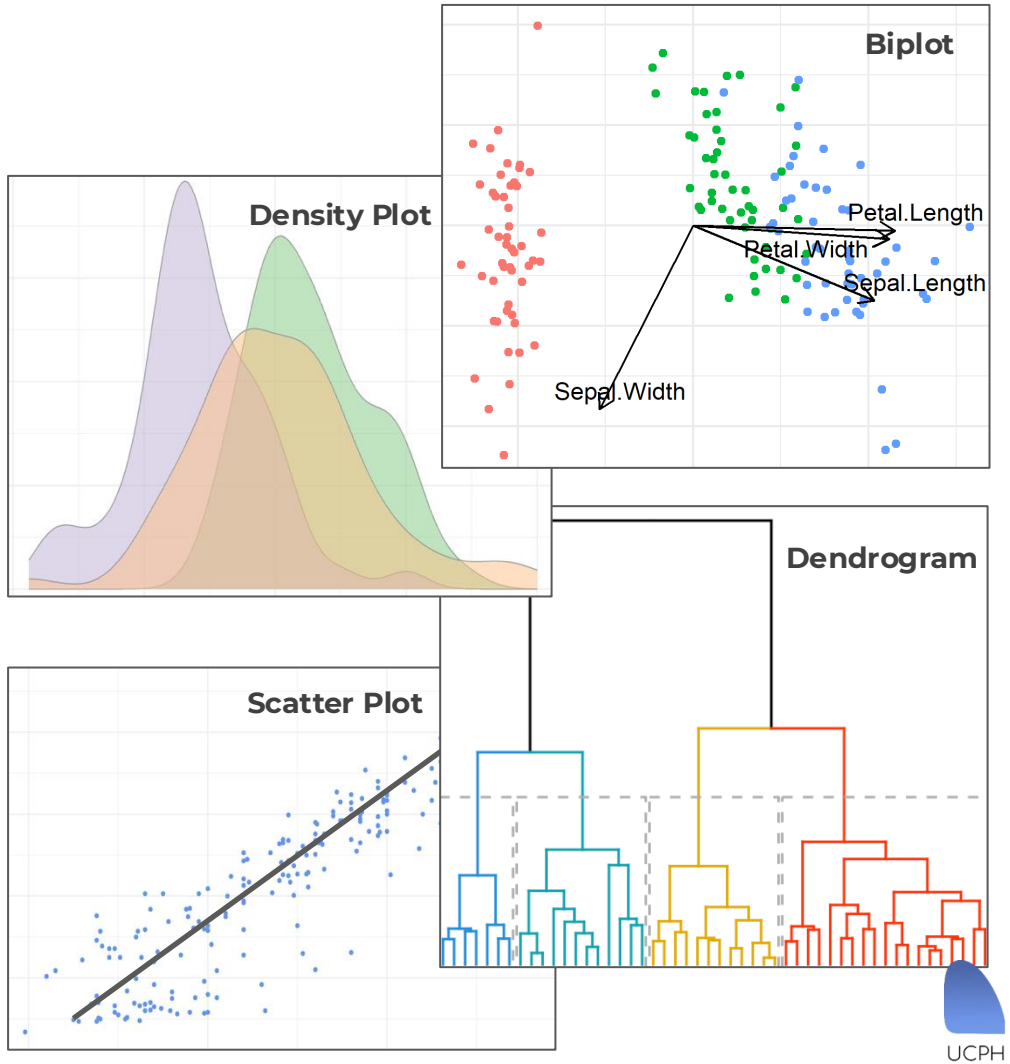




EXPLORATORY DATA ANALYSIS

Before we can start modelling we need to familiarize ourselves with the data.

- Does it look as you expected?
- Key characteristics.
- What kind of analysis can we do?
- Missing values
- Extreme outliers
- How do we need to prepare the data for analysis?

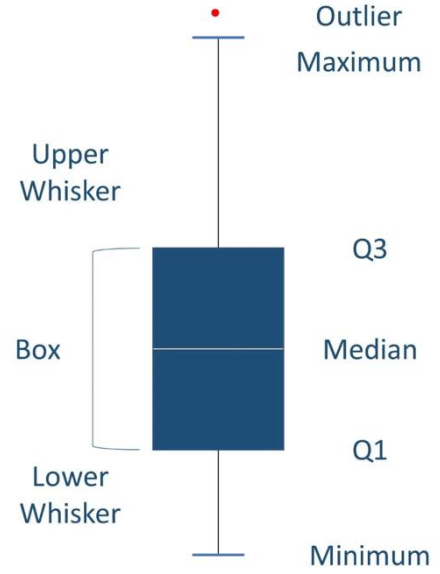


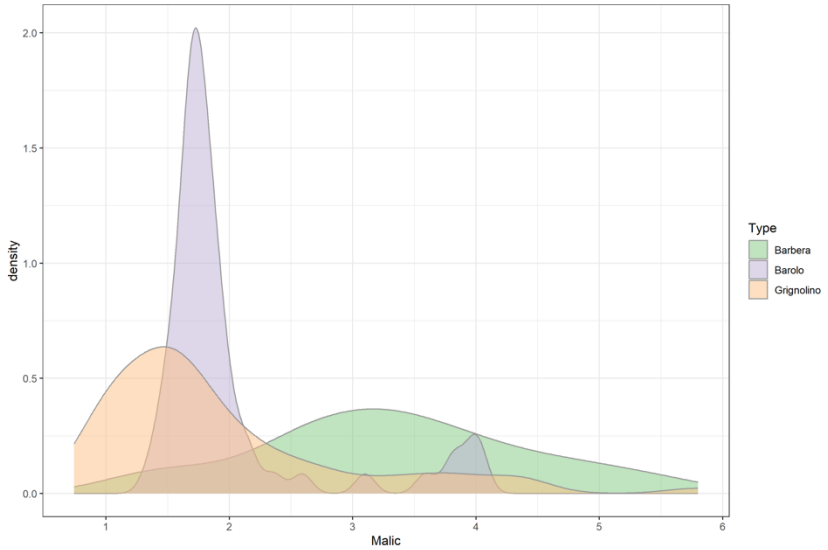
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SUMMARY STATISTICS

Summary statistics are used to distill a dataset into some key characteristics.

- Central tendency (mean, median, mode)
- Spread (standard deviation)
- Minimum and Maximum
- Quartiles
- Shape of the distribution
- Correlation between features





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DATA CLEANING AND SET-UP

We want to clean up our data:

- Remove data outliers or duplicates
- Change one type of variable to another type
- Merge, add or remove variables

Missing values:

- Filter out some/all missing values
- Impute missing values (Regression, Maximum Likelihood, ML)

Set-up:

- Object type (array, dataframe, dictionary, etc.)
- Layout, long VS wide format

Wide w. missing values

Treatment	Age	height	weight
A	55.0	NA	65.7
B	31.0	172.0	69.4
C	39.0	161.0	NA

Imputation

Treatment	Age	height	weight
A	55.0	170.0	65.7
B	31.0	172.0	69.4
C	39.0	161.0	58.1

Wide to long

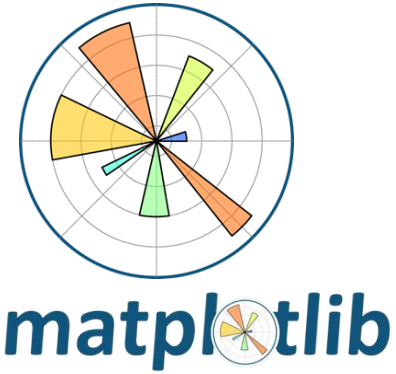
Treatment	variable	value
A	Age	55.0
A	height	170.0
A	weight	65.7
B	Age	31.0
B	height	172.0
B	weight	69.4
C	Age	39.0
C	height	161.0
C	weight	58.1

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EDA Notebook





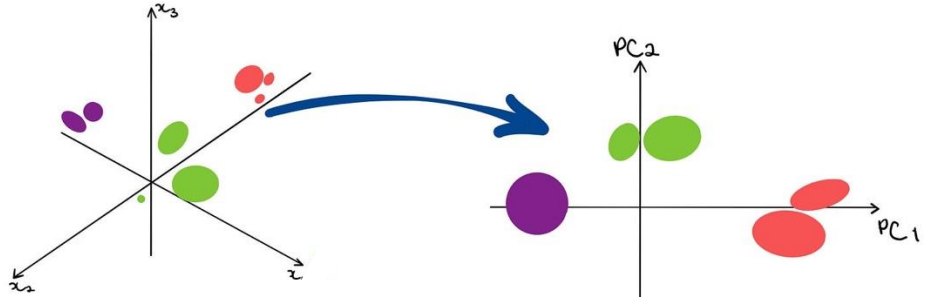




DOES THE DATA LOOK AS EXPECTED?

- Single variables may be checked by their **distribution**, (density plot) but what about **high-dimensional data**?

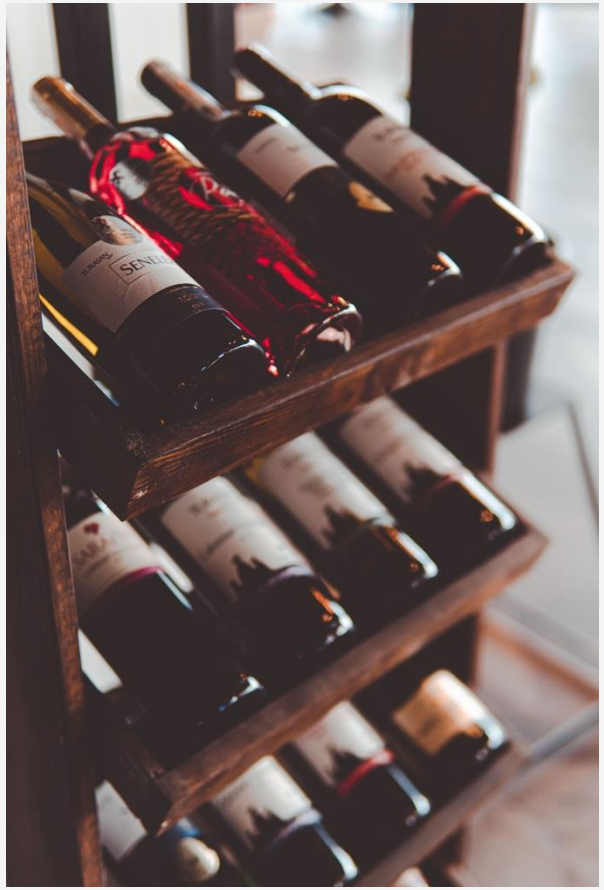
- One way to inspect the structure of your data is with a **Principal Component Analysis (PCA) Plot**



- PCA is a **dimensionality reduction technique**. It linearly transforms data from the original high dimensional space into a low (2 or 3) dimensional space, which the human eye can view and interpret.

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PRINCIPAL COMPONENT ANALYSIS



Collection of wine bottles as an example. We have measured 13 different features such as alcohol content, color, alkalinity, and flavonoids.

PCA lets us derive a set of **new dimensions** that best describe all the original wine features while also being much lower dimensional.

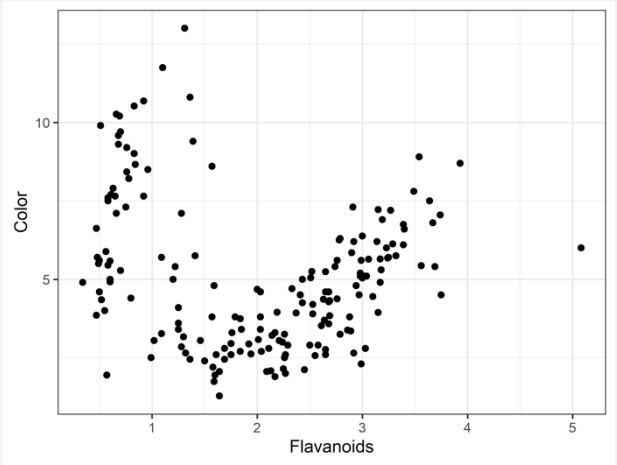
These derived dimensions are the **principal components** and they will help us understand the structure of our data.

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WHAT HAPPENS DURING PCA?

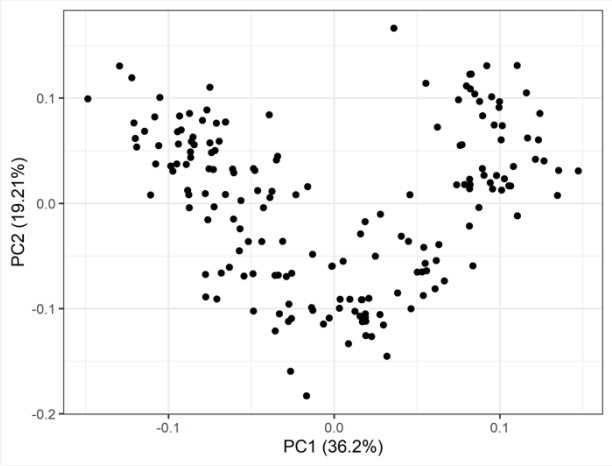
Original data with n = 13 dimensions

Wine type	Alc	Alkalinity	Flavonoids	Color	...
Barolo	14.38	12.0	3.64	5.4	...
Barolo	12.93	18.0	2.41	4.5	...
Grignolino	11.66	16.0	1.57	3.8	...
...	...	...	...	...	...



Mapped into PCA space with 2-4 dimensions

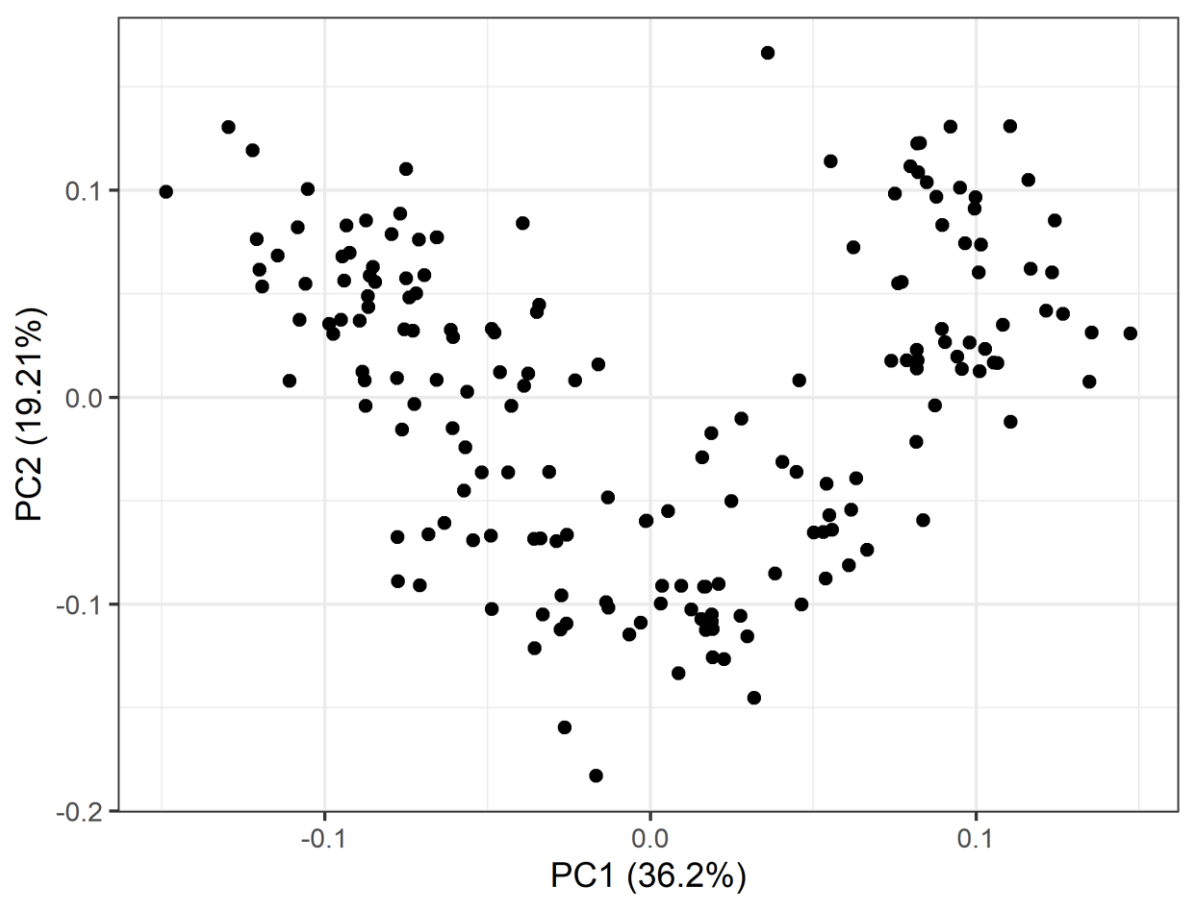
Wine type	PC1	PC2	PC3	PC4
Barolo	-3.3	1.4	-0.2	0.2
Barolo	-2.2	-0.3	-2.0	0.3
Grignolino	1.5	-1.4	-0.8	2.8
...	...	...	...	...



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PRINCIPAL COMPONENT ANALYSIS

We can now plot all wine bottles in our data by the values of their first two PCs:

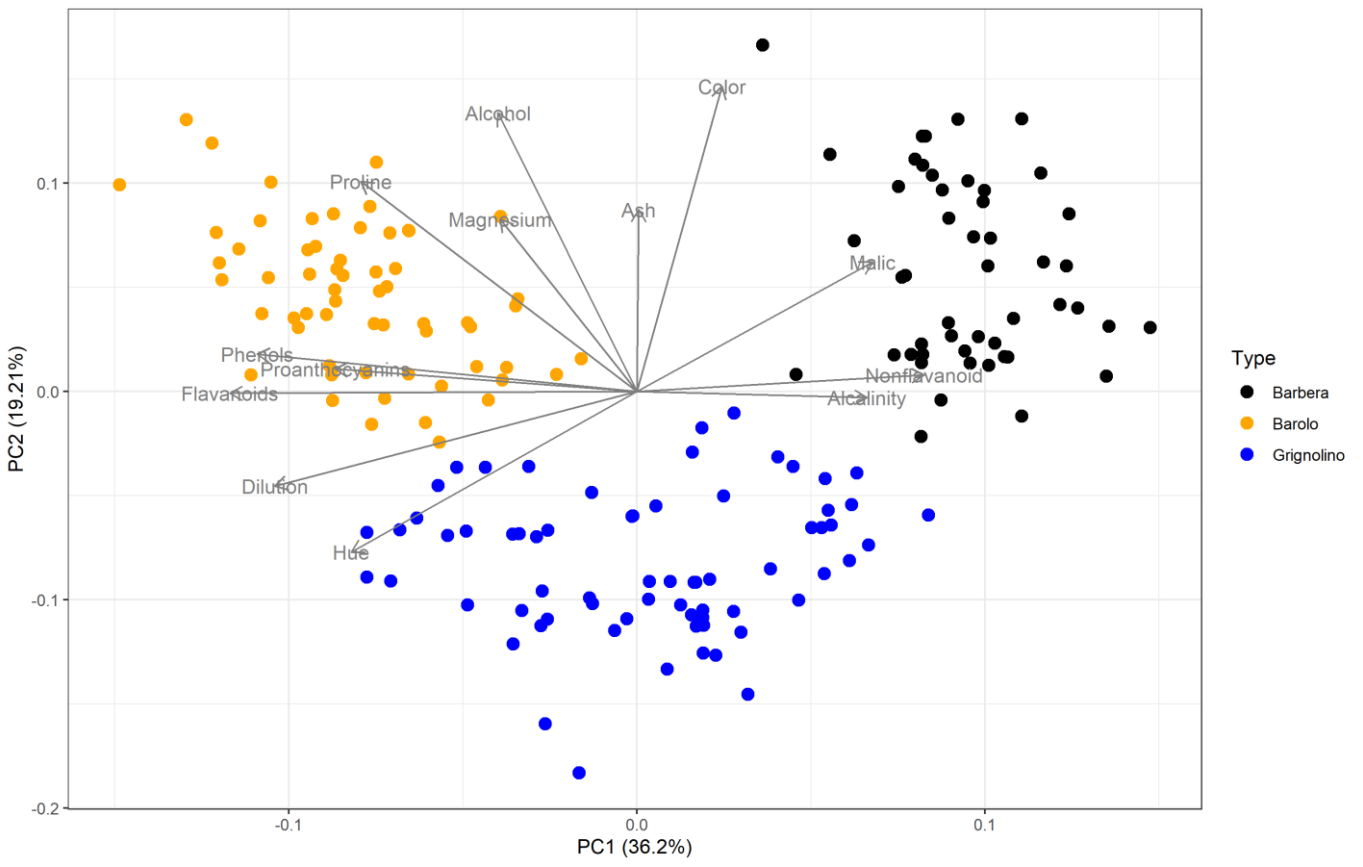


- Each dot is one wine bottle.
- If two wines are close in their PC1 and PC2, they are similar in the original feature space.
- This helps us to visually understand the structure of our data (we cannot plot in 13 dimensions!).

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PRINCIPAL COMPONENT ANALYSIS

Coloring by wine type reveals clusters!



If clusters are well separated there is a combination of original features that **explains** which cluster a data point belongs to.

We can use this to understand what defines each cluster and predict which cluster a new data point belongs to.

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WHAT HAPPENS DURING PCA?

Formally PCA is a singular value decomposition such that:

Original data, Scaled and centered

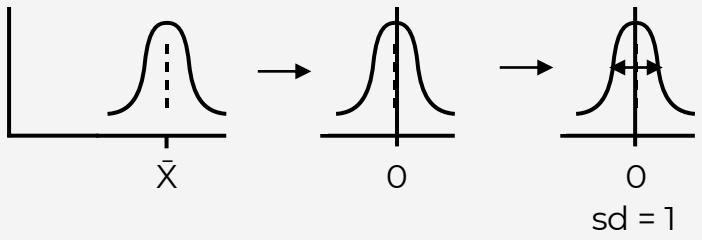
Data projected into the PC space *- the scale*

Eigenvectors (weights) *- the direction*

$$\mathbf{X} = \tilde{\mathbf{U}}\mathbf{S}\mathbf{V}^{\top}$$

178 wines x 13 features

Alc	Alkalinity	Flavonoids	Color	...
1.51	-1.17	1.03	0.25	...
0.25	-2.48	0.73	-0.29	...
1.50	-0.96	0.97	0.57	...
...	...	...	...	...



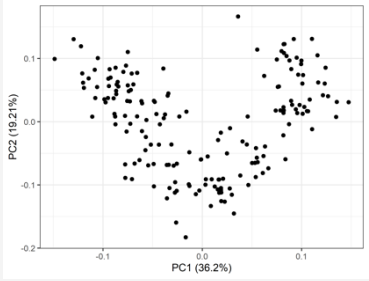
178 wines x 4 PCs

PC1	PC2	PC3	PC4
-3.3	1.4	-0.2	0.2
-2.2	-0.3	-2.0	0.3
1.5	-1.4	-0.8	2.8
...	...	...	...

4 PCs x 13 features

PC	Alc	Alkalinity	Flavonoids	Color	...
PC1	-0.14	0.24	-0.42	0.08	...
PC2	0.48	-0.01	0.01	0.53	...
PC3	-0.21	0.61	0.15	-0.14	...
PC4	0.02	-0.06	-0.15	-0.07	...

These are the dots in a PCA plot.



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WHAT HAPPENS DURING PCA?

Every PC is a linear combination of all original dimensions:

PC1 value for bottle 1

Alcohol value for bottle 1

Alkalinity value for bottle 1

Flavonoid value for bottle 1

Color value for bottle 1

$$PC1_{b1} = e_{alc,PC1} * alc_{b1} + e_{alk,PC1} * alk_{b1} + e_{flav,PC1} * flav_{b1} + e_{col,PC1} * col_{b1} \ldots$$

Eigenvector of PC1 in *Alcohol*

Eigenvector of PC1 in *Alkalinity*

Eigenvector of PC1 in *Flavonoids*

Eigenvector of PC1 in *Color*

Scaled and centered data

Alc	Alkalinity	Flavonoids	Color	...
1.51	-1.17	1.03	0.25	...
0.25	-2.48	0.73	-0.29	...
1.50	-0.96	0.97	0.57	...
...	...	...	...	...

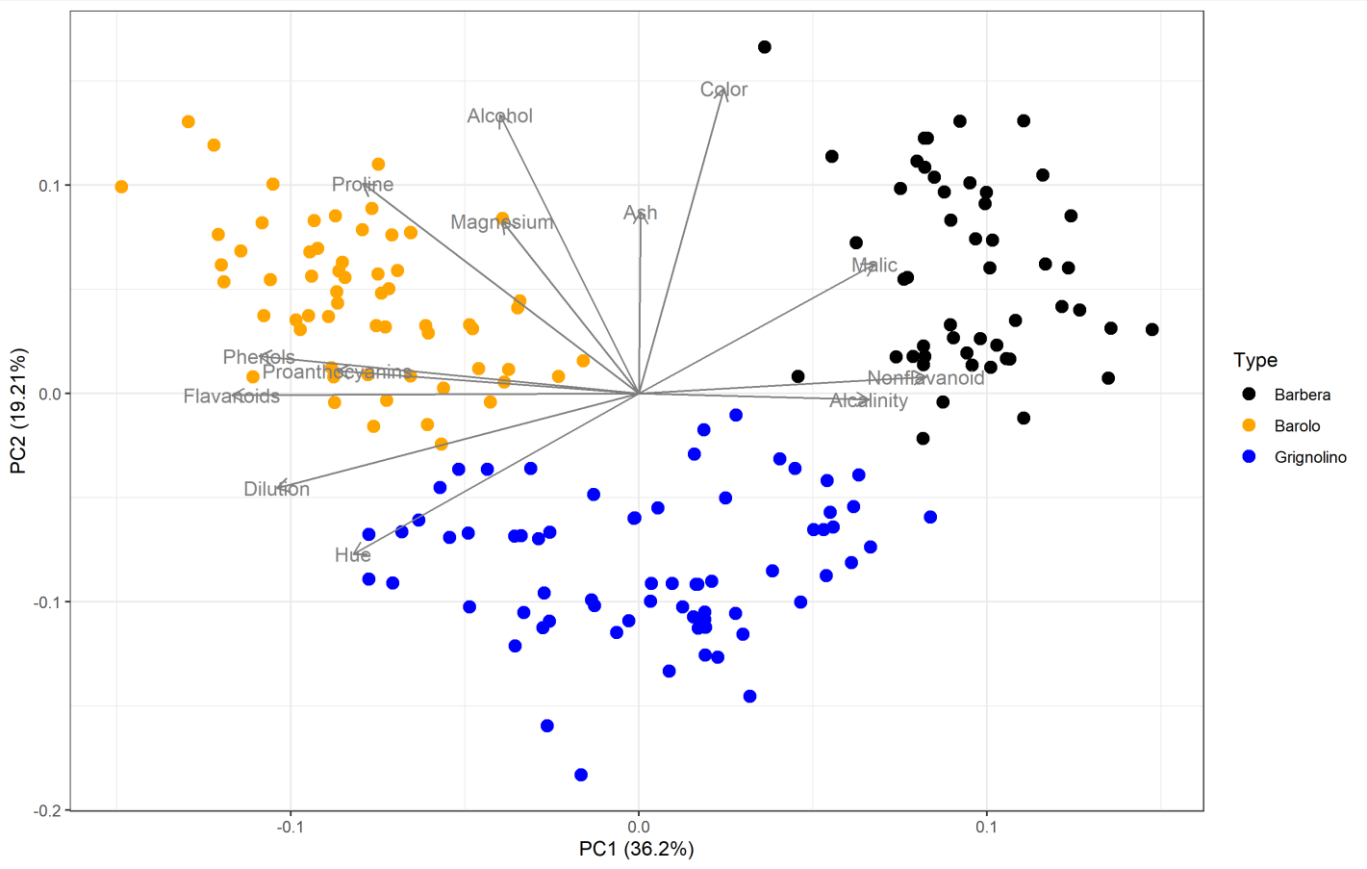
4 PCs x 13 features

PC	Alc	Alkalinity	Flavonoids	Color	...
PC1	-0.14	0.24	-0.42	0.08	...
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PC4	0.02	-0.06	-0.15	-0.07	...

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PRINCIPAL COMPONENT ANALYSIS

We can also project the Eigenvectors into the plane of the PCs.



This helps us to visually interpret how each of the original dimensions contributes to each PC.

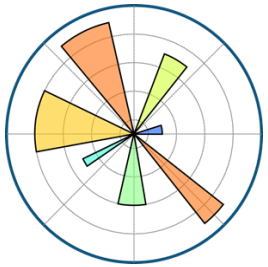
Transposed Eigenvectors:

PC	Alc	Alkalinity	Flavonoids	Color	...
PC1	-0.14	0.24	-0.42	0.08	...
PC2	0.48	-0.01	0.01	0.53	...
PC3	-0.21	0.61	0.15	-0.14	...
PC4	0.02	-0.06	-0.15	-0.07	...

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PCA Notebook



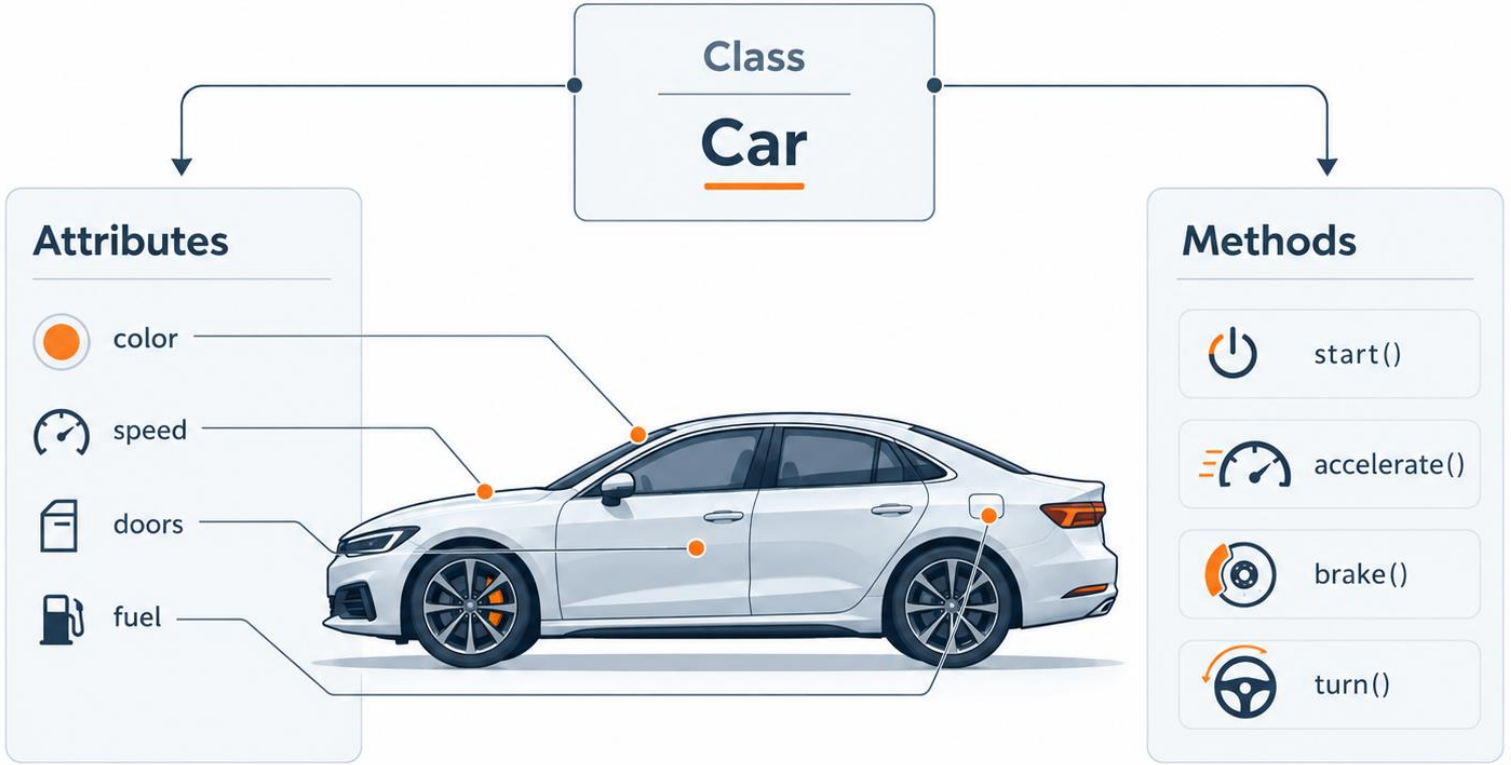


matplotlib





Classes, attributes and methods



Python II:

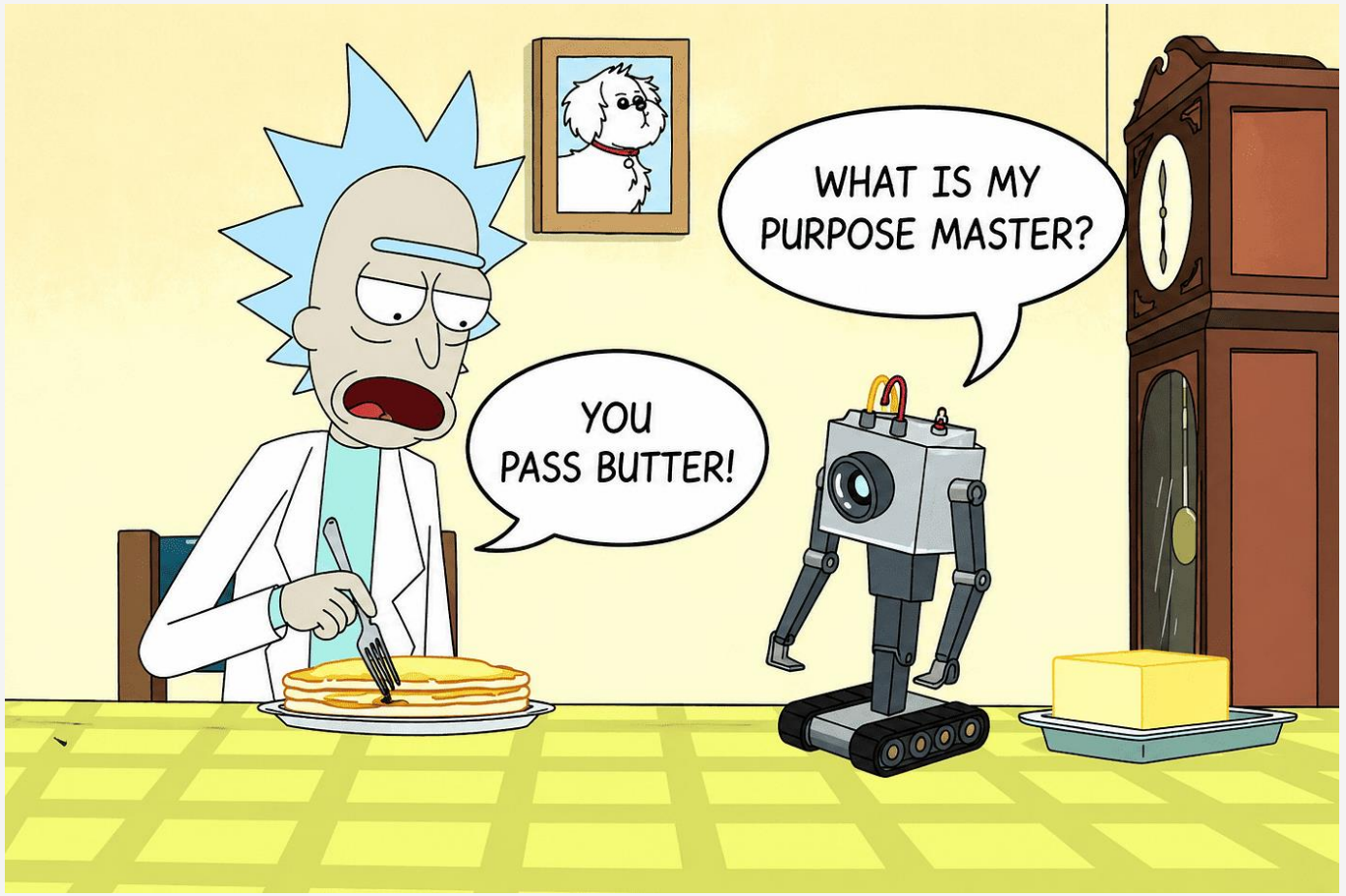
Data Analysis in Python

DAY 2



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Functions



WHERE WE LEFT OFF



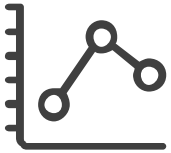
DATA COLLECTION



EXPLORATORY DATA ANALYSIS



MODELLING



MODEL EVALUATION





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DATA ANALYSIS

The outcome is known

The outcome is unknown
or there are no labels

Supervised Learning

Unsupervised Learning

Regression

Classification

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MODELLING

We are specifically interested in a **mathematical** model that describes the relationship between predictors/features $\mathbf{x_i}$ and the outcome **y**. This helps us to:

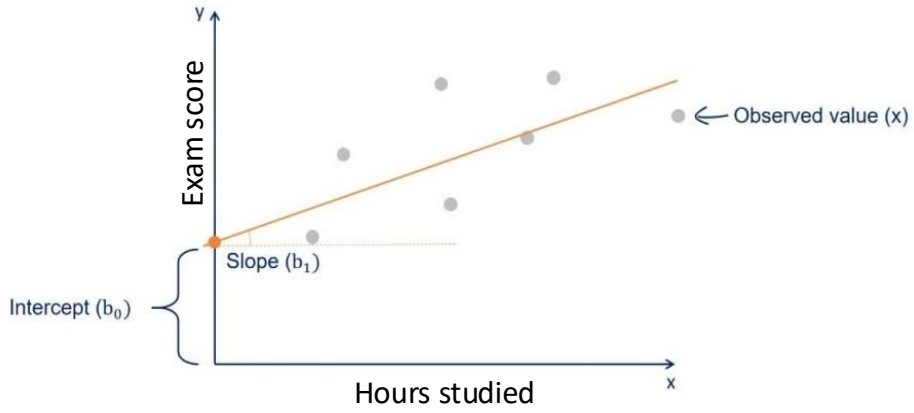
- Quantify the predictive power of **x** over **y**
- Qualify the relationship (is it linear, ect)
- Predict new **y**'s from known values of $\mathbf{x_i}$

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MODELLING - REGRESSION

1. Pick a model
2. Fit the model parameters to our data
3. Evaluate model performance
4. Interpret model parameters

$$\hat{y}_i = b_0 + b_1 x_i$$



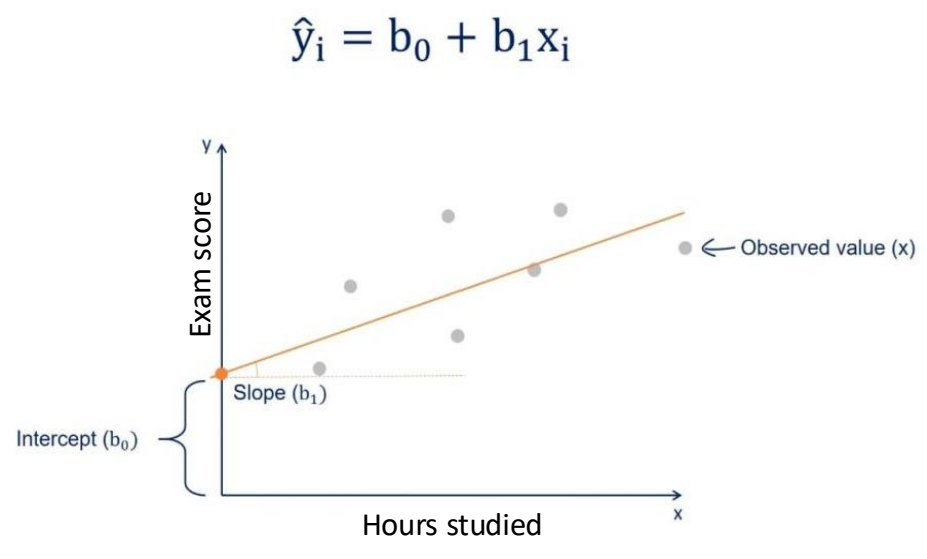
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MODELLING - REGRESSION

1. Pick a model

Linear Regression:

The exam score (**y**) can be approximated as a base value (intercept $\mathbf{b_0}$) + the hours studied (**x**) * a coefficient (slope $\mathbf{b_1}$).



MODELLING - REGRESSION

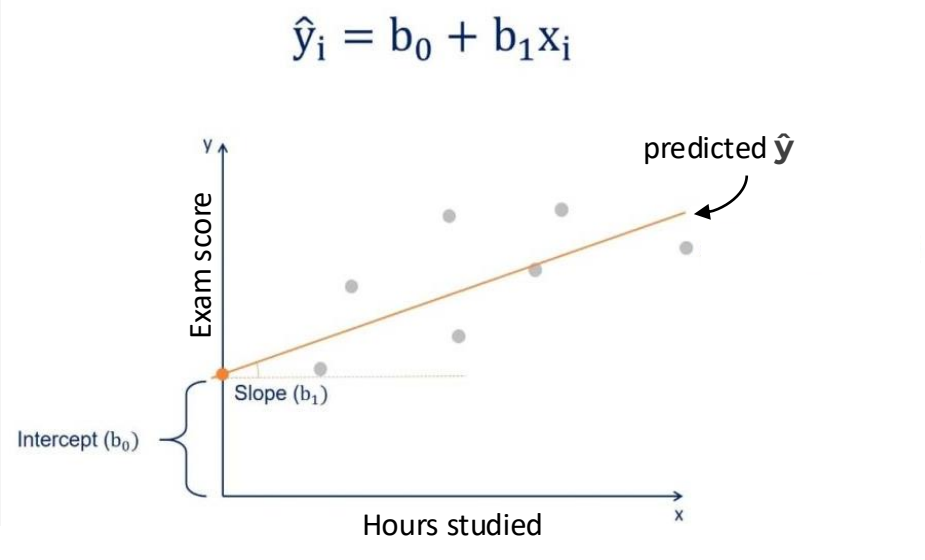
2. Fit the model

Our hypothesis is that

$\hat{y} = b_0 + b_1 * x$

where $\hat{y}$ is the predicted outcome.

We want to find b_0 and b_1 such that $\hat{y}$ is as similar as possible to y. b_0 and b_1 are also referred to as the model's trainable parameters.



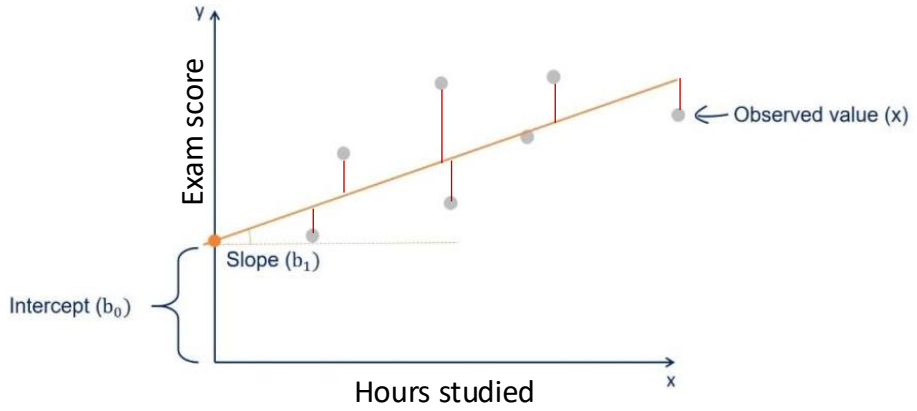
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MODELLING - REGRESSION

3. Evaluate model performance

Now that we have a model with fitted parameters we need to quantify the discrepancy between **ŷ** and **y.** This will be our model performance.

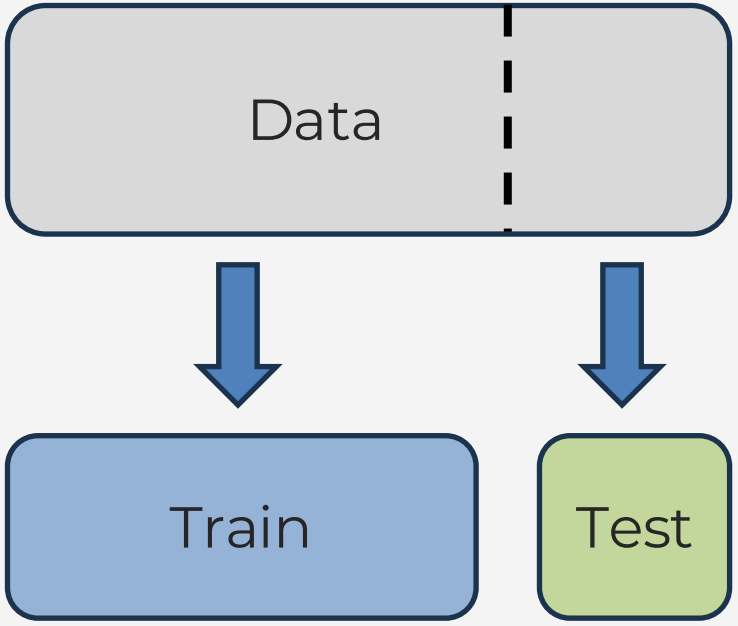
$$\hat{y}_i = b_0 + b_1 x_i$$



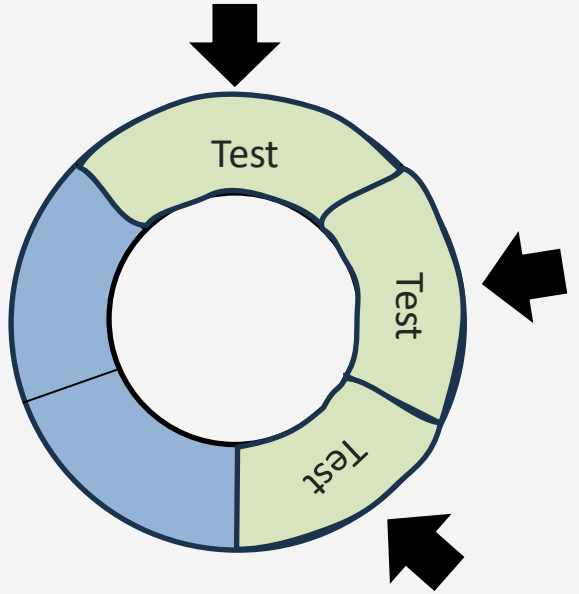
$$MSE = \frac{1}{n}\sum\underbrace{\left(y - \hat{y}\right)^2}_{\text{The square of the difference between actual and predicted}}$$

TRAIN AND TEST DATA

In order to have a fair evaluation of model performance, we only use part of the data for training (fitting) and reserve another (small) part to test performance.



Cross Validation



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MODELLING - REGRESSION

4. Interpret model parameters

Our linear regression model has this form:

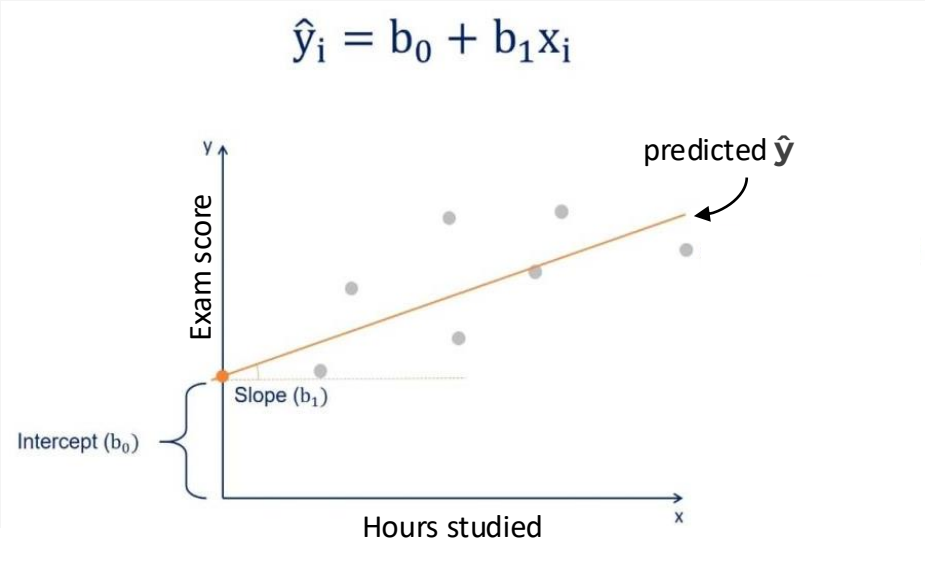
$\hat{y} = b_0 + b_1 * x$

And we have estimated parameters b_0 and b_1.

What do they mean?

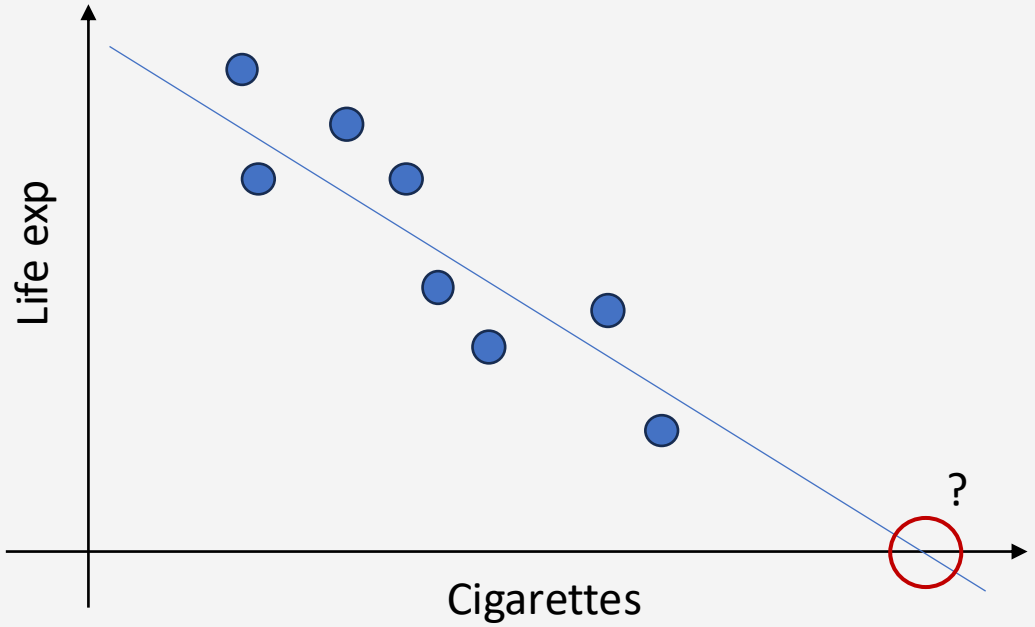
b_0 – the predicted value of $\hat{y}$ when **x** is 0

b_0 – the predicted change in $\hat{y}$ for a one unit change in **x**



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Linear regression

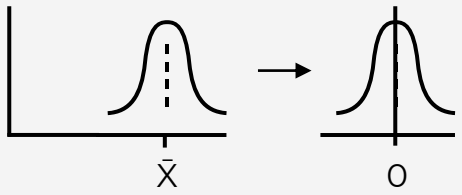


CENTERING PREDICTORS

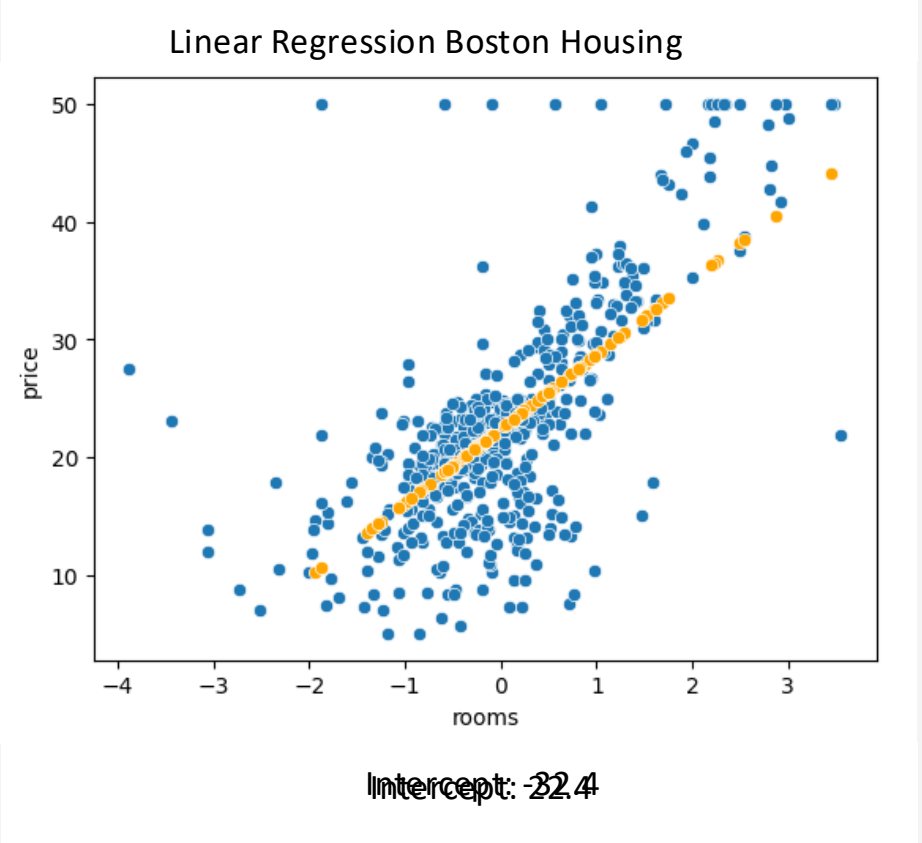
The intercept is the outcome value we get when the predictors are at 0. But some predictors do not include 0 in their range so then the intercept doesn't make much sense.

Fix:

Center the predictor.



Now the intercept is the outcome we expect when the predictor takes its **mean** value. This does **not** change the predictive performance!

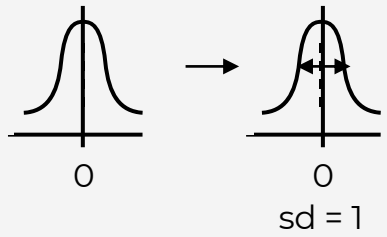


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SCALING PREDICTORS

Another technique that is sometimes employed is scaling the predictors (features).

This means moving them from whichever scale they are on to a scale with a std dev of 1.



This also does **not** change the predictive performance! But it has implications for the interpretation:

$$\hat{y} = b_0 + b_1 * x_1 + b_2 * x_2$$

Size of product is constrained

Change in life exp. estimate for each extra daily cigarette.

Change in life exp. estimate for each extra minute of exercise.

x_1 the number of cigarettes smoked

x_2 the number of minutes of exercise

$\hat{y}$ the life expectancy

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REGRESSION WITH CATEGORICALS

Our example model:

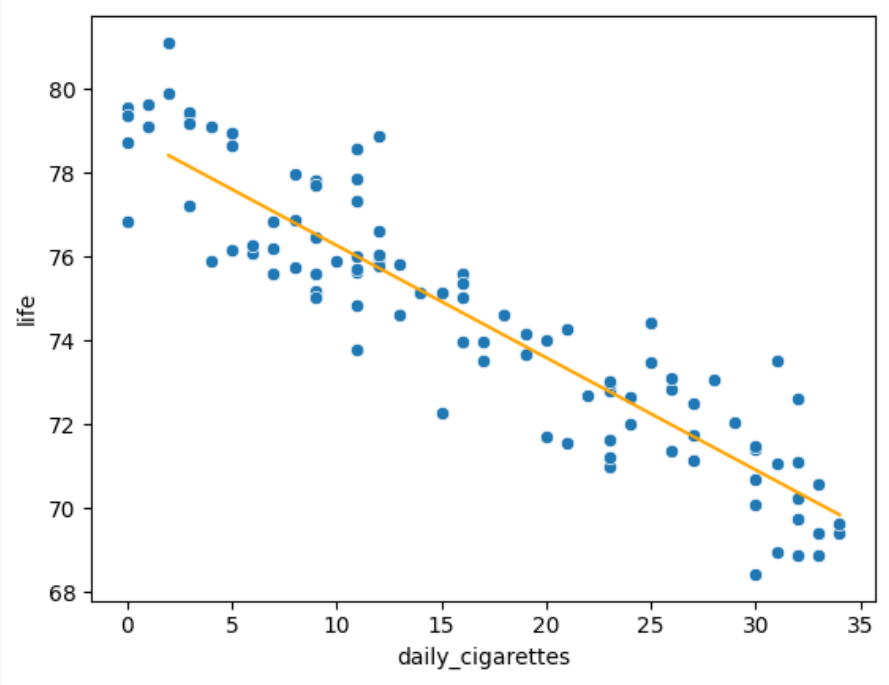
$\hat{y} = b_0 + b_1 * x$

With estimated parameters:

life ~ 78.4 + (-0.26) * daily cigarettes

This allows us to model the life expectancy as a base value (intercept) + the number of daily smoked cigarettes times a coefficient b_1. We can conceivably add other numerical predictors and estimate their coefficients.

But how can we incorporate **categorical** predictors?



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REGRESSION WITH CATEGORICALS

Let's say we have a binary predictor 'exercise' we can use a model of this form:

life = intercept + coef_c * daily_cigs + coef_e * exercise with **exercise ∈ {0,1}**

Or in more math terms:

$$\hat{y} = b_0 + b_1 * x_1 + b_2 * x_2$$

Since $x_2 \in \{0,1\}$ we have only two cases:

$\hat{y} = b_0 + b_1 * x_1 + b_2 * x_2$ if $x_2 == 1$ (the person exercises)

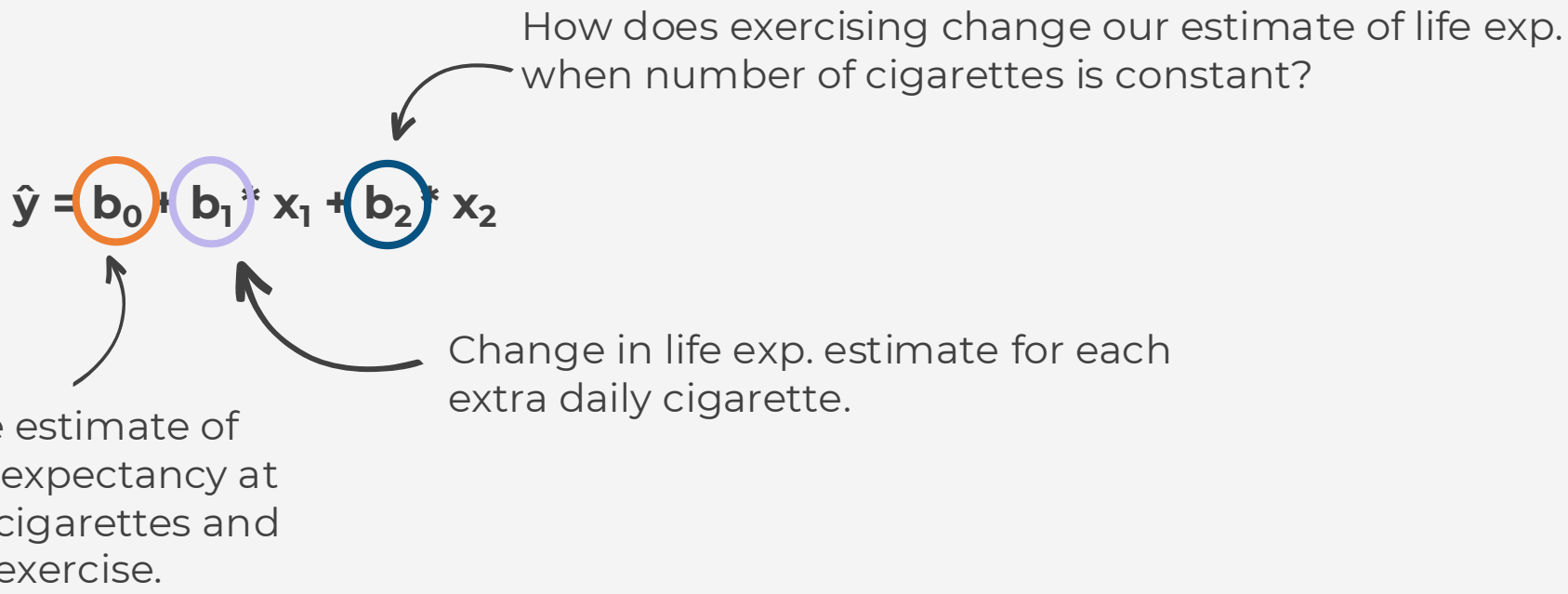
$\hat{y} = b_0 + b_1 * x_1$ if $x_2 == 0$ (the person does not exercise)

For our example we therefore expect to find $b_1 < 0$ and $b_2 > 0$ and all is well.

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REGRESSION WITH CATEGORICALS

We now have 3 parameter estimates:



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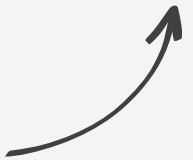
REGRESSION WITH CATEGORICALS

What happens if the predictor has more than two levels?

life = intercept + coef_c * daily_cigs + coef_e * exercise

with **exercise ∈ {0,1,2,3}**

Levels coded as integers



We can still do:

$$\hat{y} = b_0 + b_1 * x_1 + b_2 * x_2$$

But mathematically we'll be saying:

- There is no effect of doing level 0 == no exercise
- A person who does level 1 exercise gains b_2 years of life expectancy
- A person who does level 2 exercise gains double as much!
- And at level 3 triple!

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REGRESSION WITH CATEGORICALS

Instead of integers we can use what is called dummy or one hot encoding:

$\hat{y} = b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$ with **exercise ∈ {0,1,2}**

Now we can say:

If exercise == 0: $x_2 = 0$ & $x_3 = 0$

If exercise == 1: $x_2 = 1$ & $x_3 = 0$

If exercise == 2: $x_2 = 0$ & $x_3 = 1$

x_2	x_3
0	0
1	0
0	1

x_2, x_3 are indicator vars that say which level of exercise we are currently modelling.

No assumption on relationship between b_2 and b_3, but 1 extra predictor so less power.

We need one less indicator var == dummy than we have levels.

And we get:

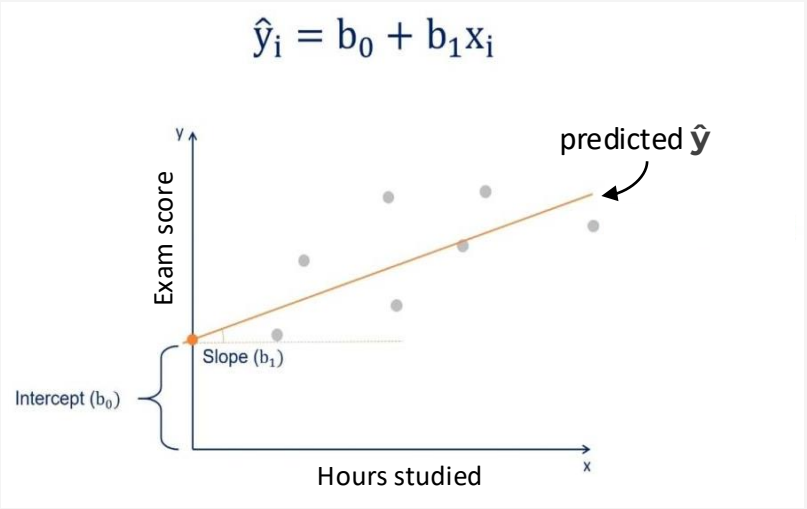
- b_2 is the gain in life exp at exercise level 1
- b_3 is the gain at level 2

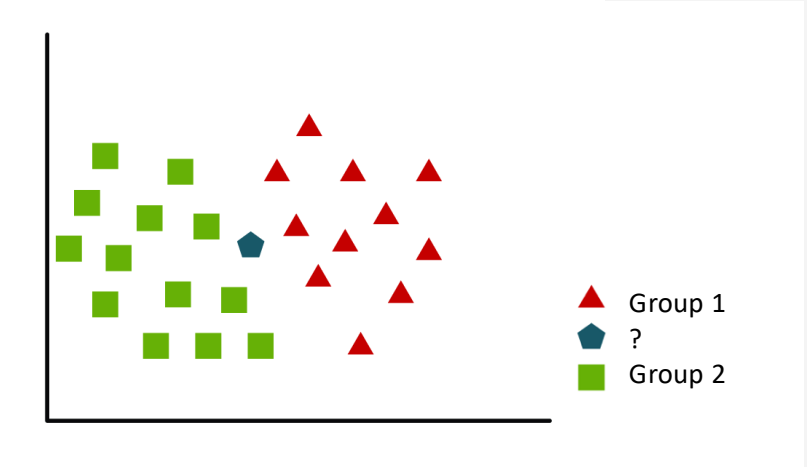
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CLASSIFICATION

When the outcome variable is binary/categorical our goal becomes classification rather than regression:

- **Regression:** Which value is a new y likely to take given we know x?
- **Classification:** Which class is a new y likely to belong to given we know x?



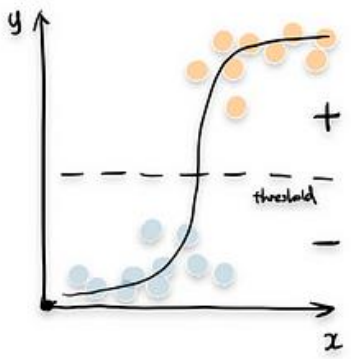


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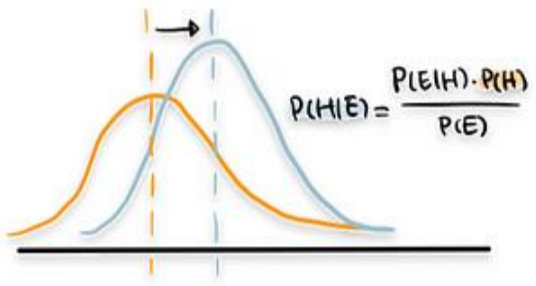
CLASSIFICATION MODELS

There are many algorithms that perform classification:

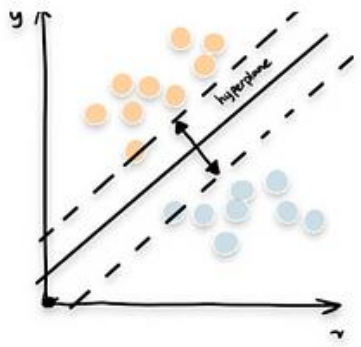
Logistic Regression



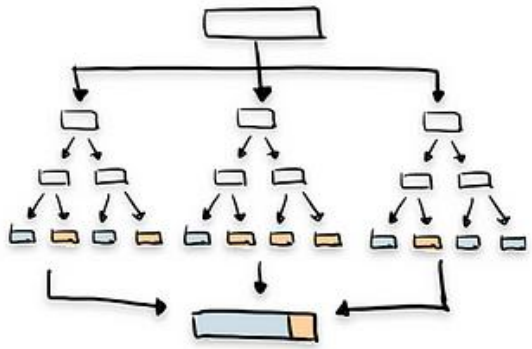
Naïve Bayes



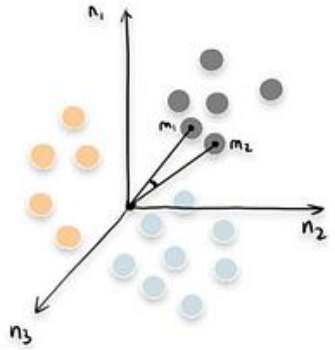
Support Vector Machine



Random Forest



K-Nearest Neighbors



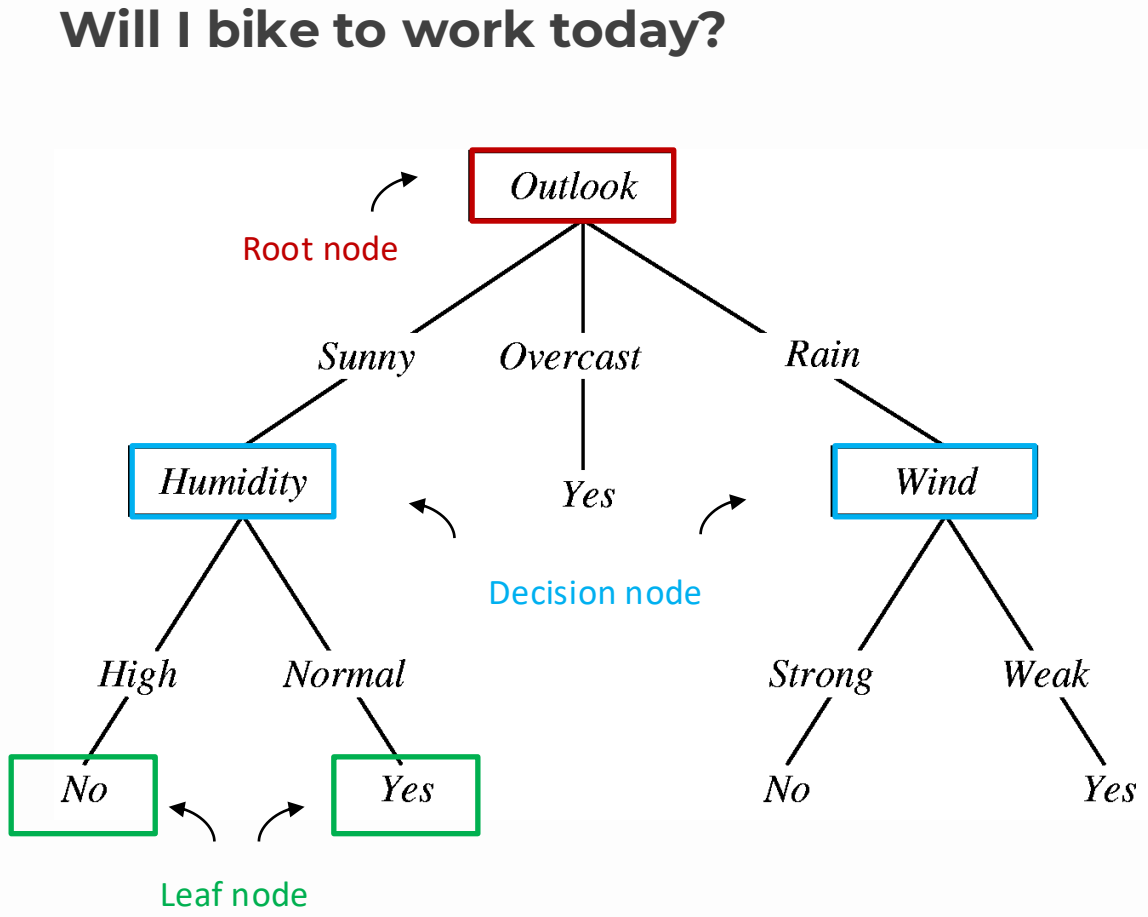
https://towardsdatascience.com/top-machine-learning-algorithms-for-classification-2197870ff501

DECISION TREE

A decision tree is a series of 'checks' we follow to determine which class a data point belongs to. Each time a check is made, the dataset is split.

It has three different kinds of nodes:

- Root node:
  - Contains **all** datapoints
  - The first check is made
- Decision node:
  - Subsequent checks
- Leaf node:
  - Labeled with of the possible classes



DECISION TREE: TRAINABLE PARAMETERS

Just like a linear regression model we fit the decision tree to our data.

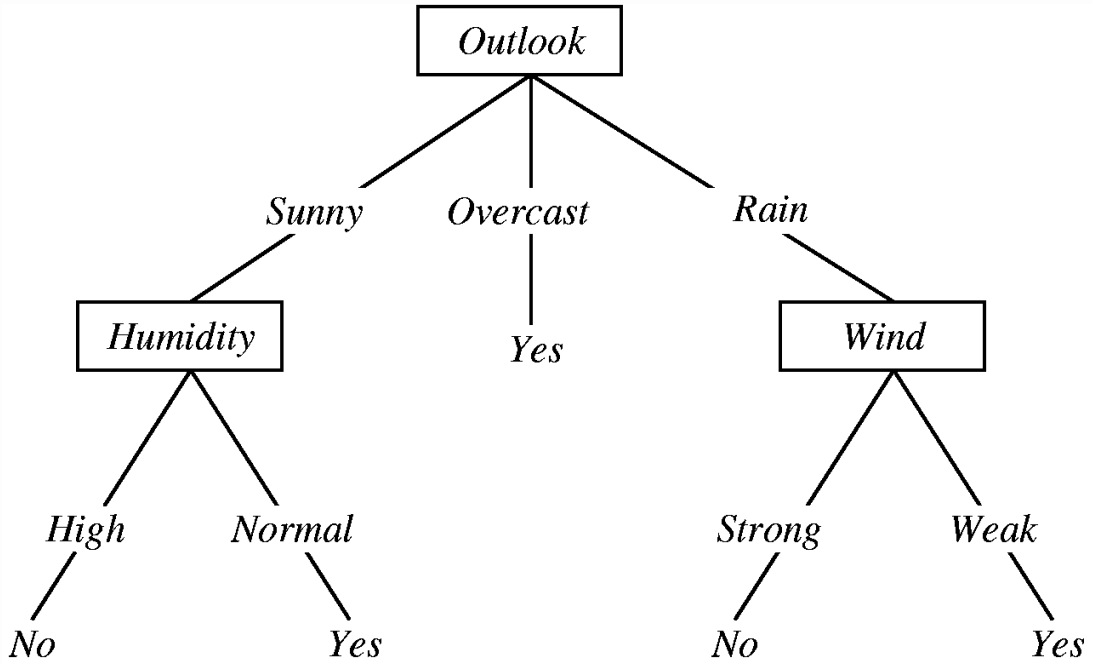
Hyperparameters:

- Criterion to evaluate the goodness of the split
- Number of leaf nodes
- Minimum amount of samples in a leaf
- Other parameters related to structure

Trainable parameters:

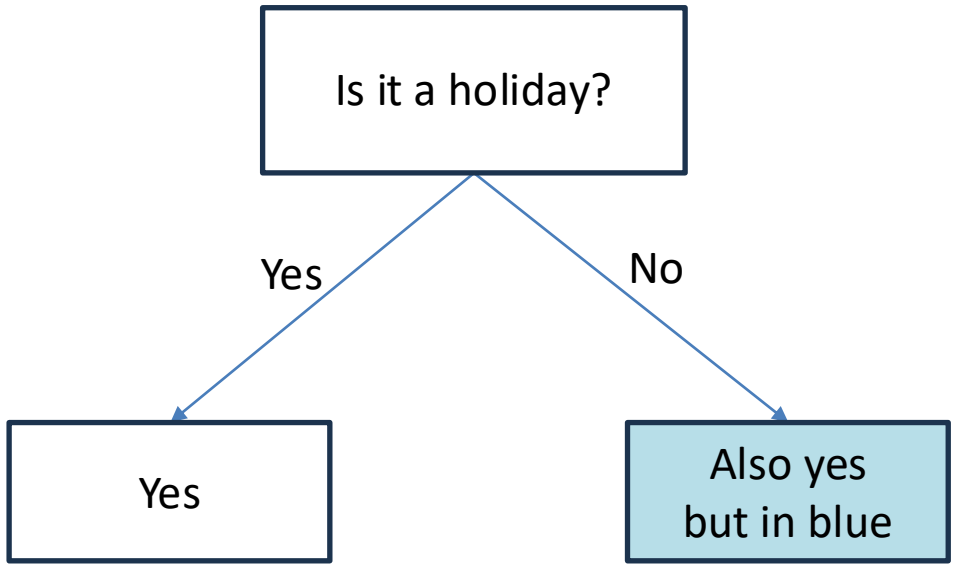
- The structure of the tree
- Feature importance

Will I bike to work today?



Decision Trees

Should I eat cake?

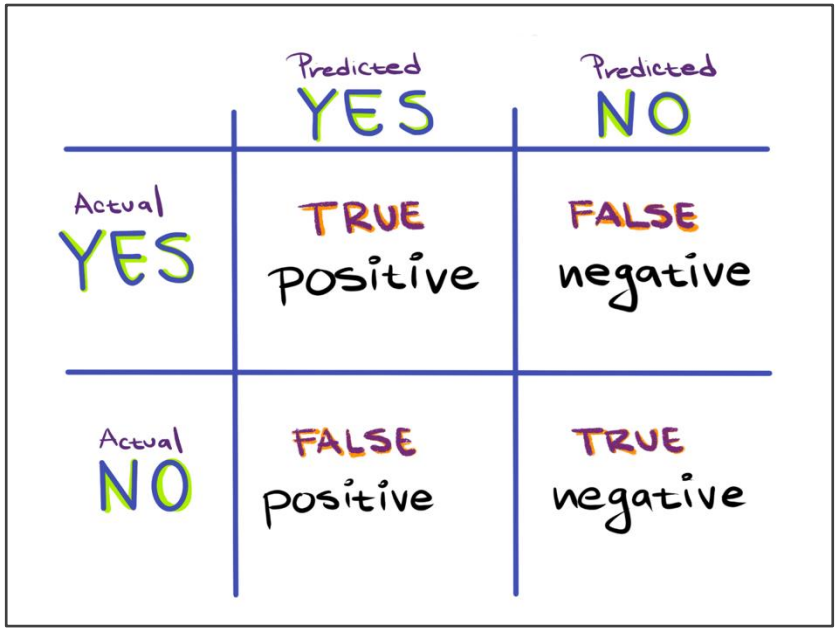


CLASSIFIER PERFORMANCE

For a classifier that output only labels, like a decision tree, we look at the numbers of True and False negatives and positives.

This is called a confusion matrix.

However ...



$$\text{Precision} = \frac{tp}{tp + fp}$$

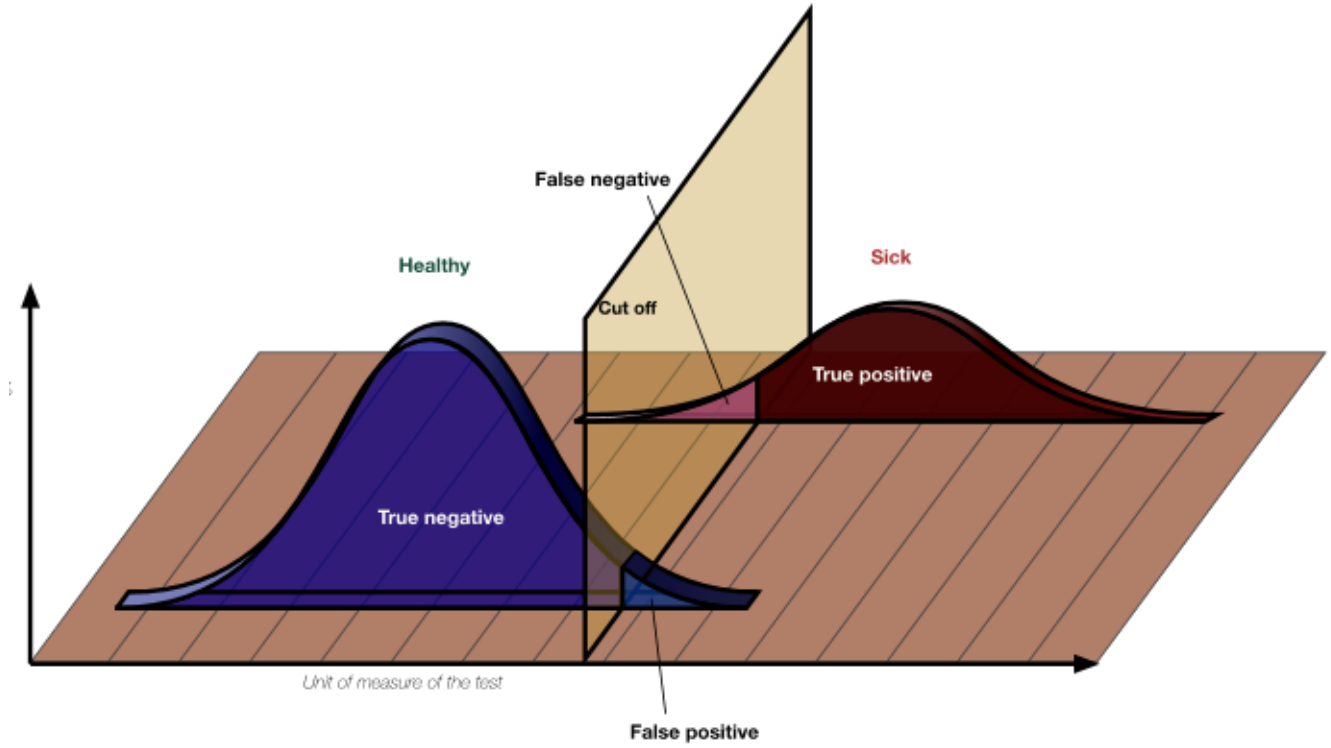
$$\text{Recall} = \frac{tp}{tp + fn}$$

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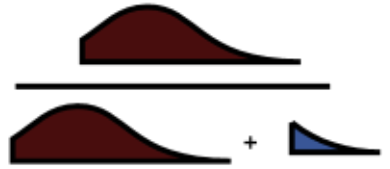
CLASSIFIER PERFORMANCE

Most other classification models have a movable decision boundary:

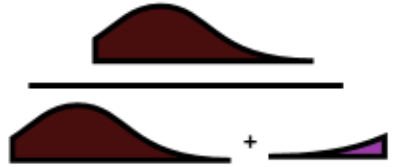
One can easily optimize on the number of false positives or false negatives, but not both at the same time. For these types of classifiers we look at the area under the curve.



Precision :



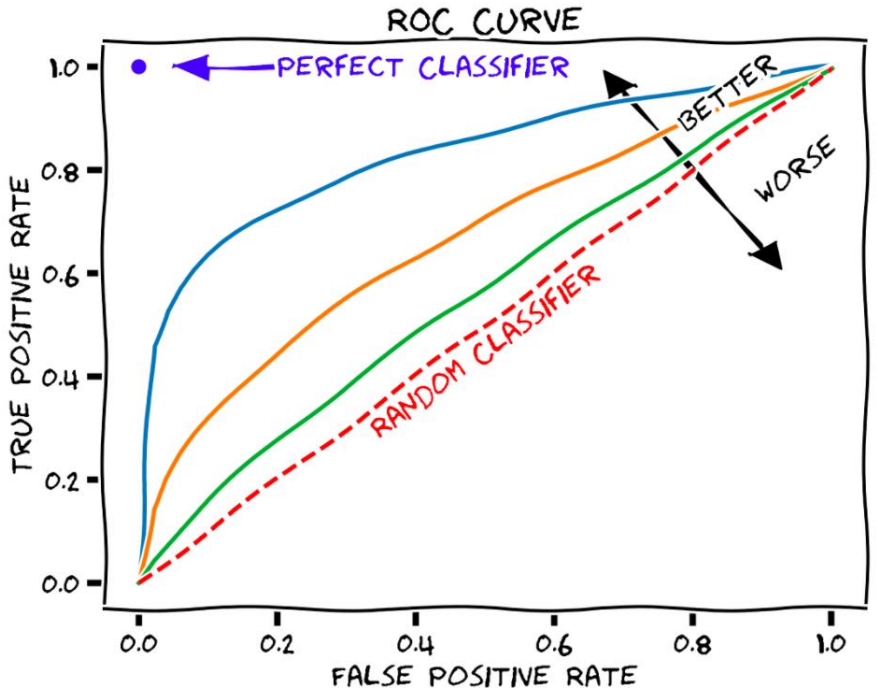
Recall/
True positive
rate



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AREA UNDER THE CURVE (AUC)

How many of the true cases have we discovered?



- Receiver Operator Curve (ROC)
- Plotted by using every possible decision boundary.
- Each True Positive Rate has an associated False Positive rate.

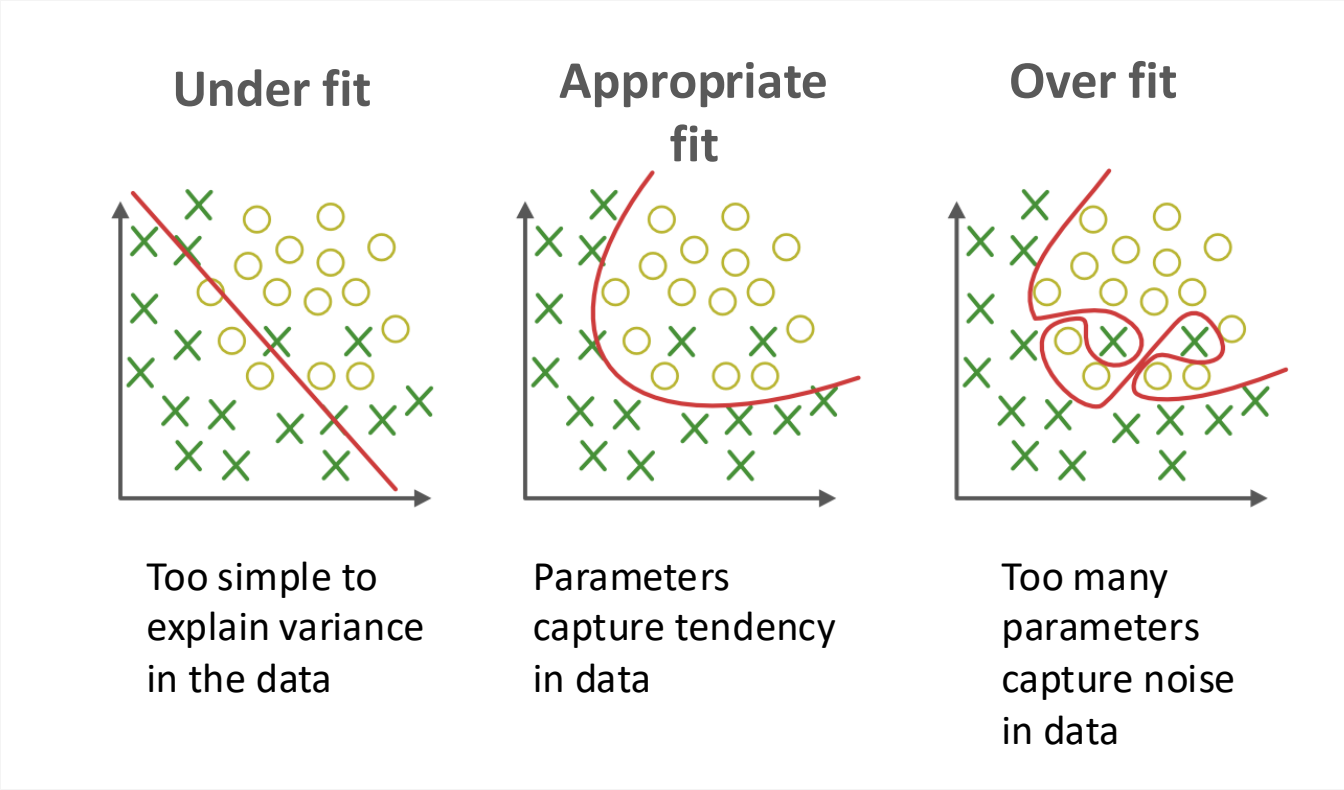
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OVERFITTING: CAPTURING NOISE

Data are inherently noisy because:

- There is no perfect measurement
- Individual differences
- Random fluctuations

A model that follows the training data too closely **generalizes poorly** to new data (with different noise) and the trends and parameters it captures may not be **meaningful**.



CIRCLING BACK

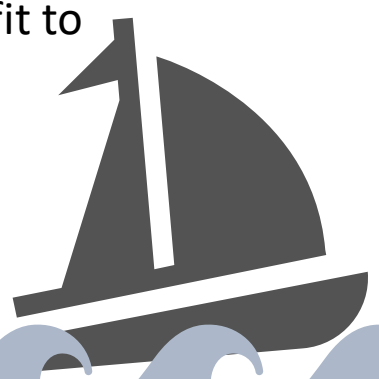
DATA COLLECTION

EXPLORATORY DATA ANALYSIS

MODELLING

MODEL EVALUATION

Prep must fit to analysis!





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Python II:

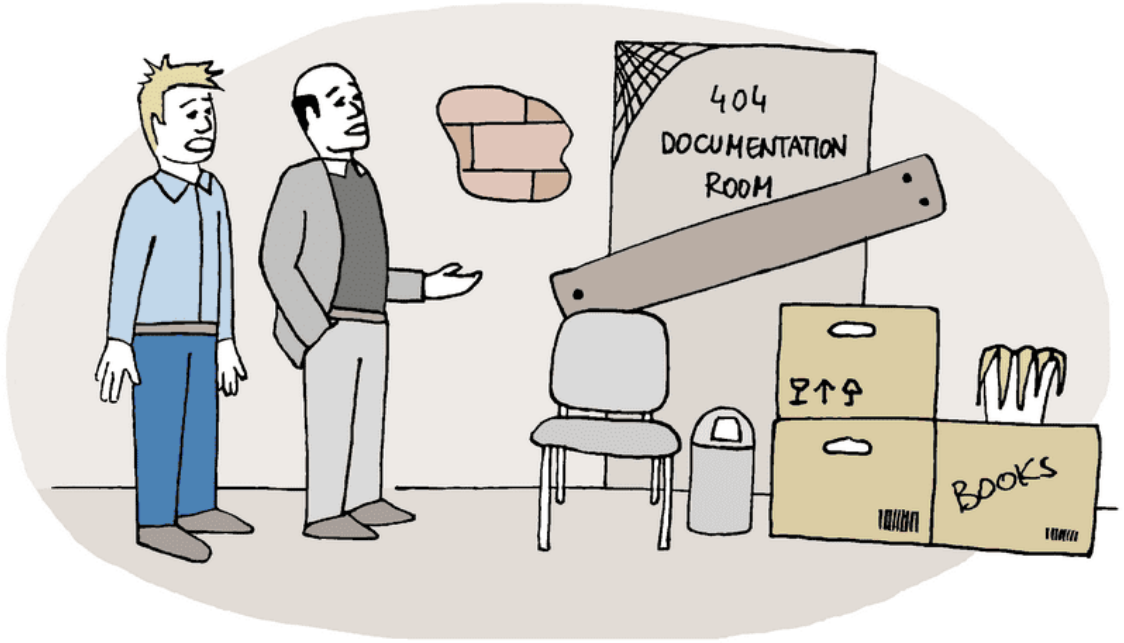
Data Analysis in Python

DAY 3



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Documentation and Testing

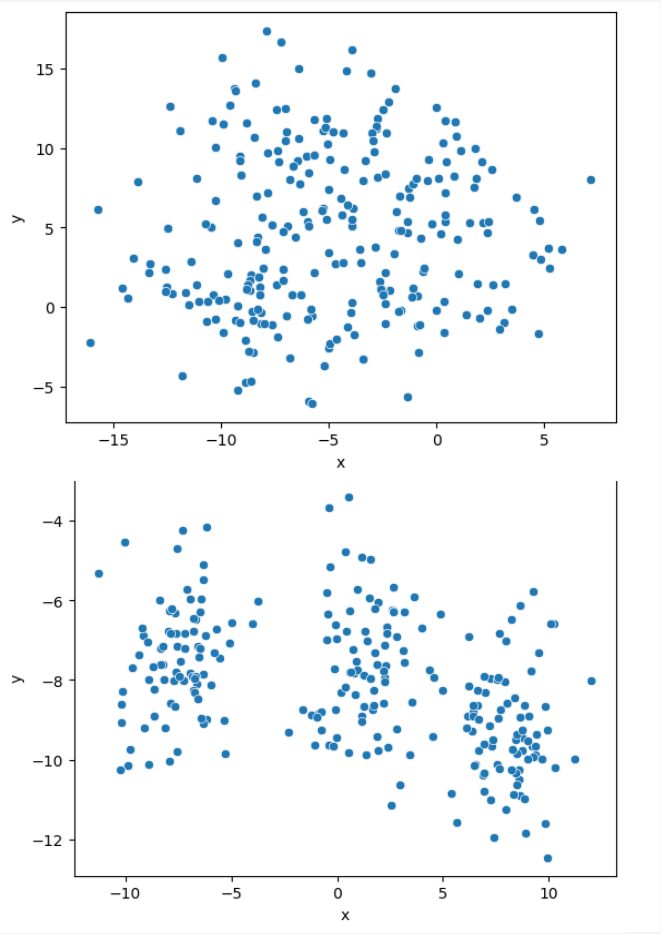


UNSUPERVISED LEARNING

What if we have data without labels? We can still do unsupervised learning to investigate questions such as:

- Are there clusters, i.e. groupings, in this dataset?
- What characterizes the members of a cluster?
- Optimal number of clusters?
- Which cluster does a new data point belong to?

We answer these questions with clustering techniques.

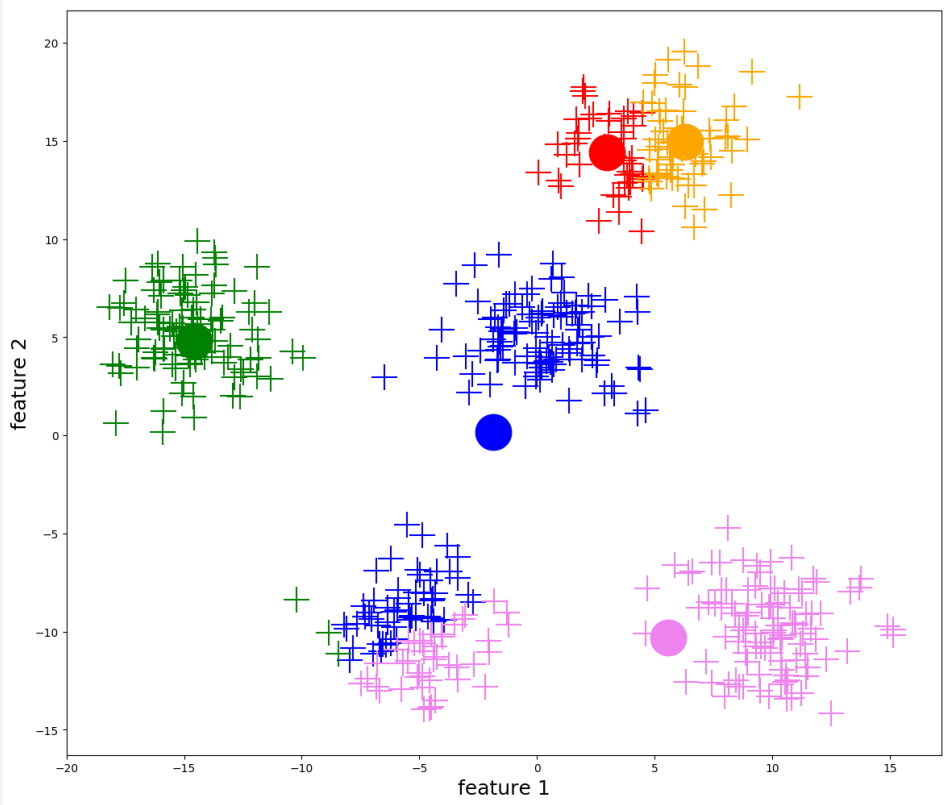


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K-MEANS CLUSTERING

K-means clustering:

1. Choose number of clusters
2. Initialize centroids
3. Assign each data point to the closest centroid
4. Recalculate centroids based on all data points that are part of the current cluster



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K-means Clustering



Classes

Class



Instances







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